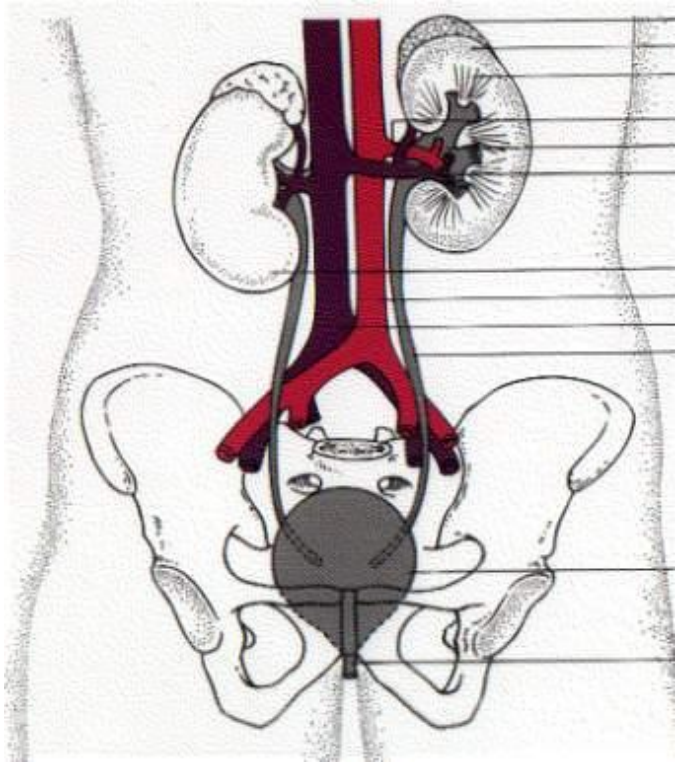
ข้างขวาต่ำกว่าซ้าย เพราะมีตับอยู่ข้างบน

ไต ⟶ ท่อไต (Ureters)

Urinary System (Renal system)



(ไต)



Human Anatomy for Nurse

อ.ดร.นาถนัดดา เกิดเอี่ยม

คณะพยาบาลศาสตร์ มหาวิทยาลัยรังสิต

วัตถุประสงค์การเรียนรู้

- บอกโครงสร้างที่สำคัญของอวัยวะใน Urinary system ได้
- บอกตำแหน่งของอวัยวะใน Urinary system ได้
- บอกหน้าที่การทำงานได้
- บอกลักษณะทางจุลกายวิภาคศาสตร์ของอวัยวะต่างๆ ใน Urinary system ได้
- บอกความแตกต่างของอวัยวะใน Urinary system ระหว่างเพศชายและเพศหญิงได้

Functions of the Urinary system

- 1. Regulating blood volume and pressure
- 2. Regulating plasma concentrations of sodium, potassium, chloride and other ions

k , Ca ≥ มีผลต่อการเต้นของหัวใจ

- 3. Stabilising blood pH
- 4. Conserving nutrients
- 5. Detoxifying poisons (with the liver)

The Urinary System

- Important functions of the kidneys
  - Maintain the chemical consistency of blood
  - Filter many liters of fluid from blood
  - Send toxins, metabolic wastes, and excess water out of the body
    - Main waste products
      - Urea
      - Uric acid
      - Creatinine

ออกรอบ ถ้าผิดปกติจะไม่เจออะไรในนี้

ก. Urea

ข. ลิโตน

ค. เม็ดเลือดแดง

↳ ถ้าผช. มี = ผิดปกติ

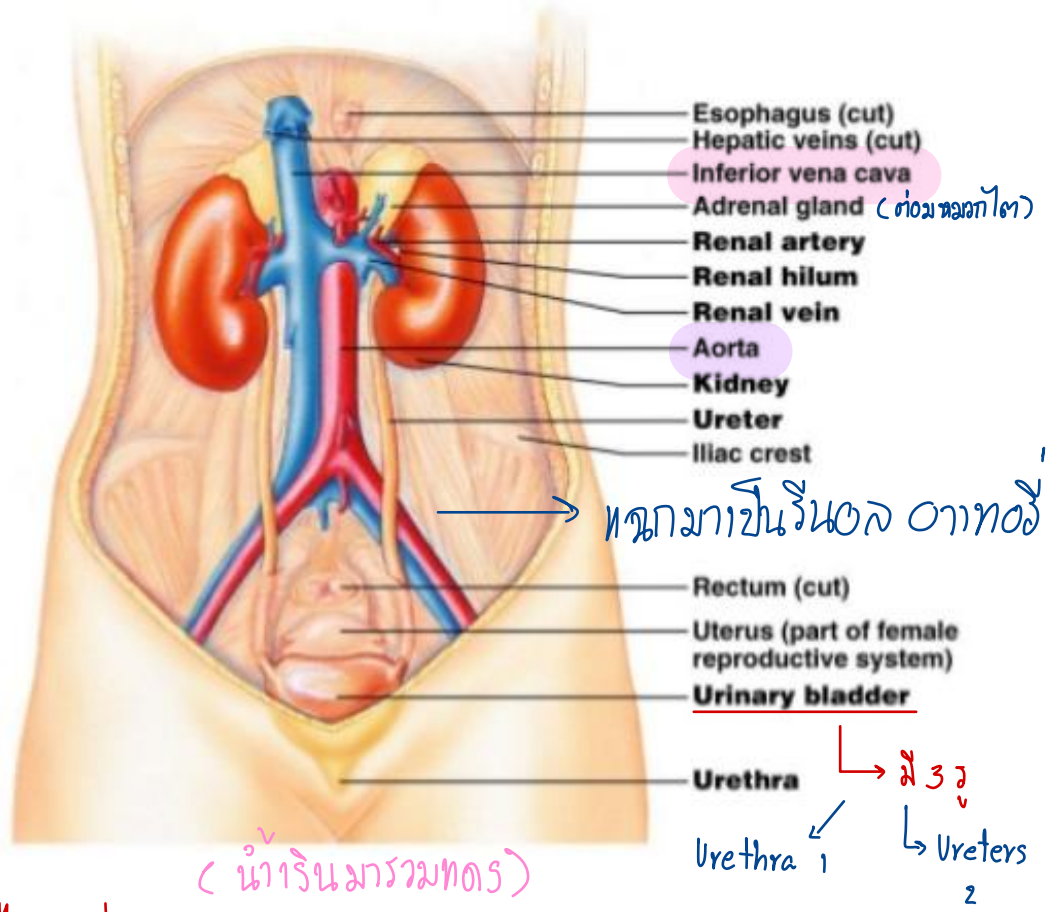
↳ ถ้าผญ. ไม่คอนเฟิร์ม = เมน

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Organs of the Urinary System

- Kidneys
- Ureters
- Urinary bladder
- Urethra

Aorta
↓
ตาเทอรี่
↓
ตาเทอริโอ้



→ หนีกมาเป็นรีนอล อาเทอรี่

มี 3 รู
Urethra 1
Ureters 2

(น้ำรีนมารวมกัน)

→ หลอดเลือดฝอย (Capilary)

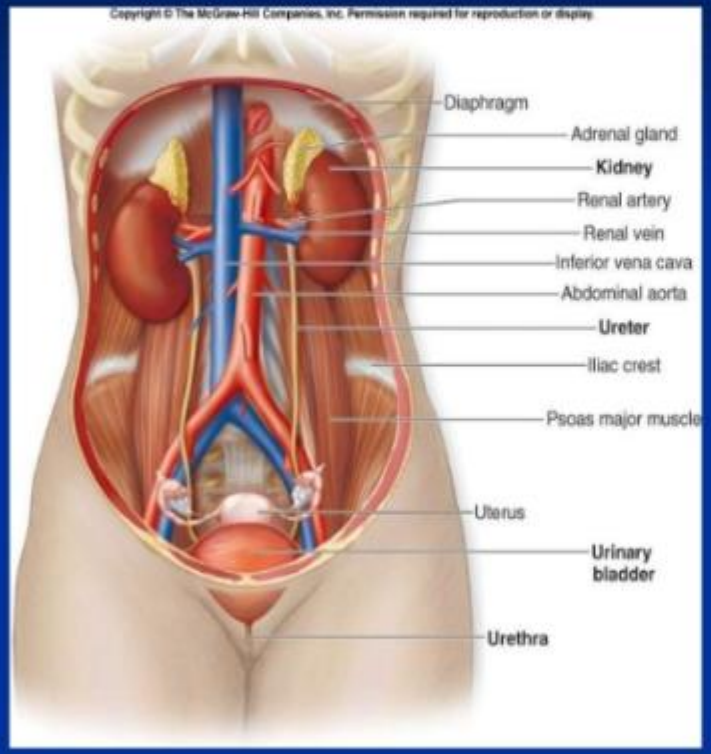
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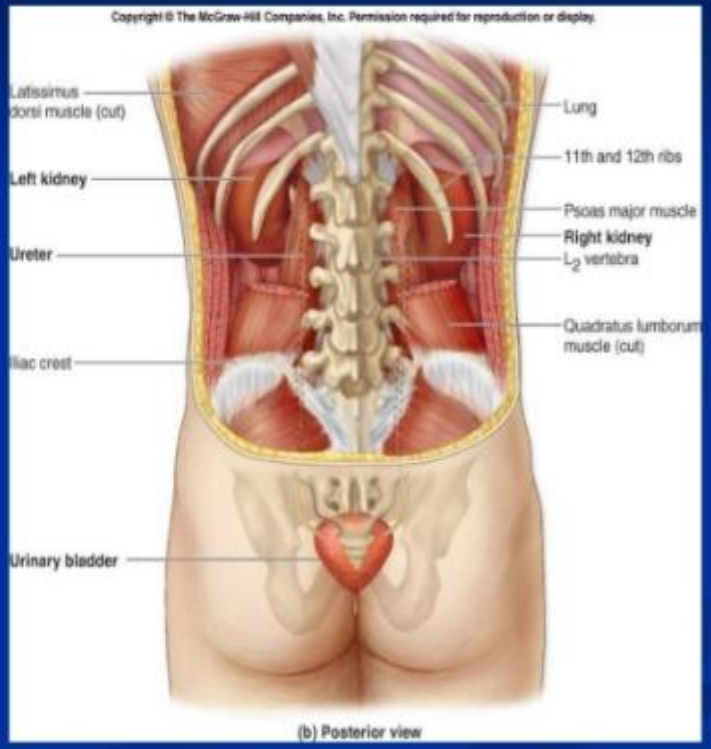
Figure 23.1a

เวนู่ → เวน → เวนาคาว่า
(เล็กสุด)

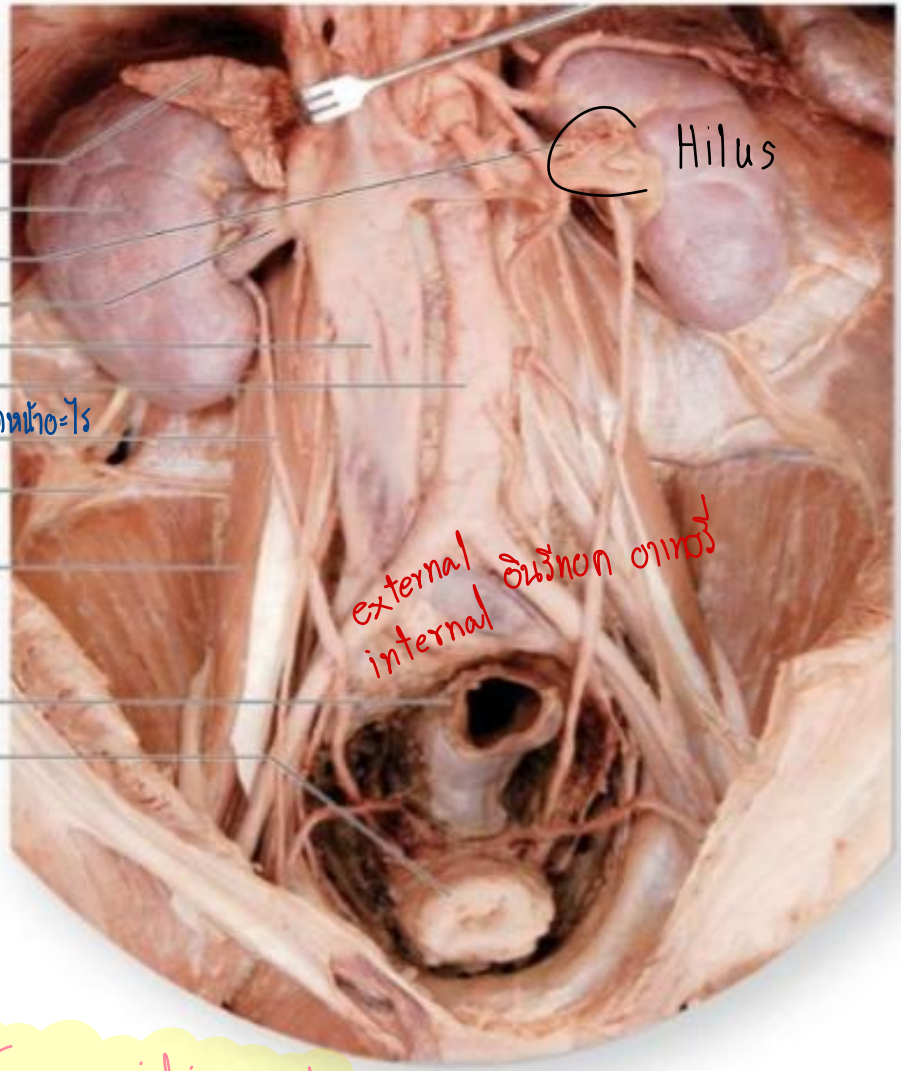
จุดที่หลกทุกอย่าง (O_2, อาหาร ฯลฯ)

Urinary System – Anterior/Posterior Views





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* ท่อรผช./ผญ. มาดหน้าอะไร

external / internal อันนี้นอก ตามชื่อ

กระเพาะปัสสาวะ

Epithelium ชนิด

↳ มี 3 ชั้น : simple
stratified
Pseudostratified

Transitional

Location and External Anatomy of Kidneys

ไปข้างหลัง

- Located **retroperitoneally**
- Lateral to T_{12}–L_3 vertebrae
- Average kidney
  - 12 cm tall, 6 cm wide, 3 cm thick
- **Hilus** (หลอดเลือดแดง แขนงเข้าไป เส้นเอ็นแขนงออกมา)
  - On concave surface
  - Vessels and nerves enter and exit
- Renal capsule surrounds the kidney

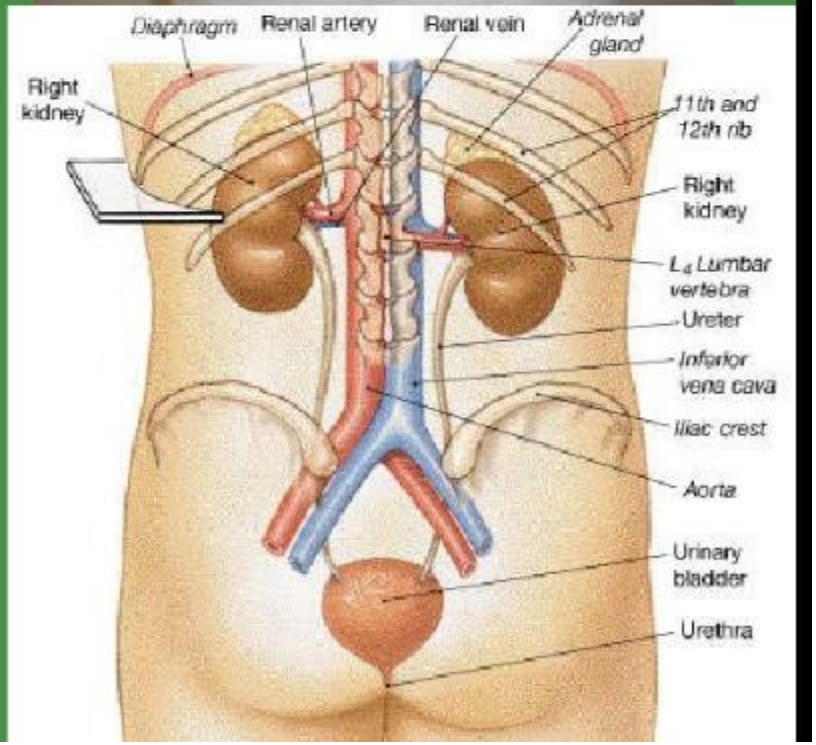
↖ มีอันนี้หุ้มอยู่

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Kidney Location

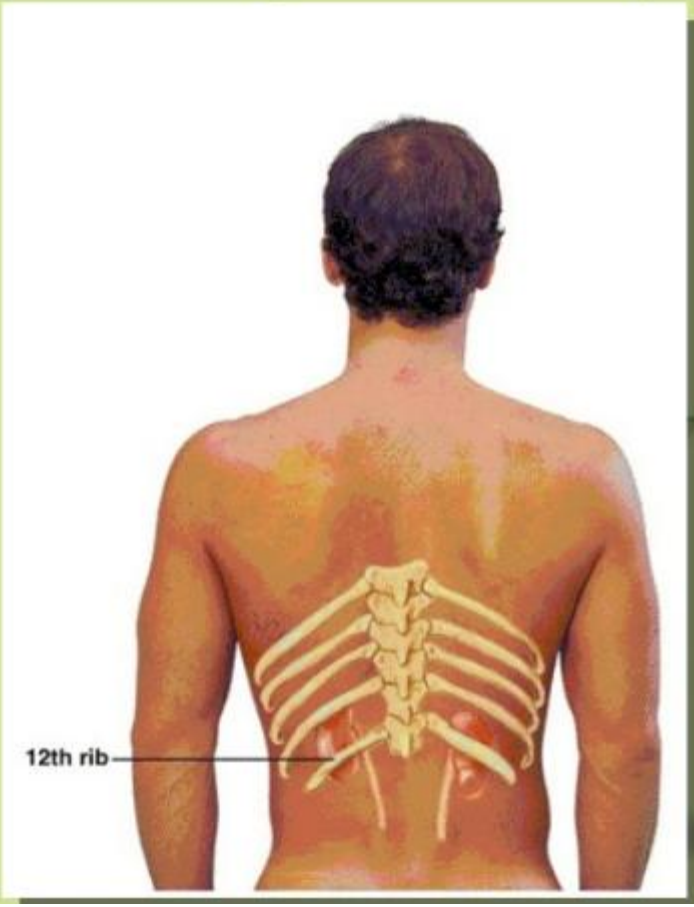


Lateral to vertebral column high on body wall, under floating ribs
in retro-peritoneal position



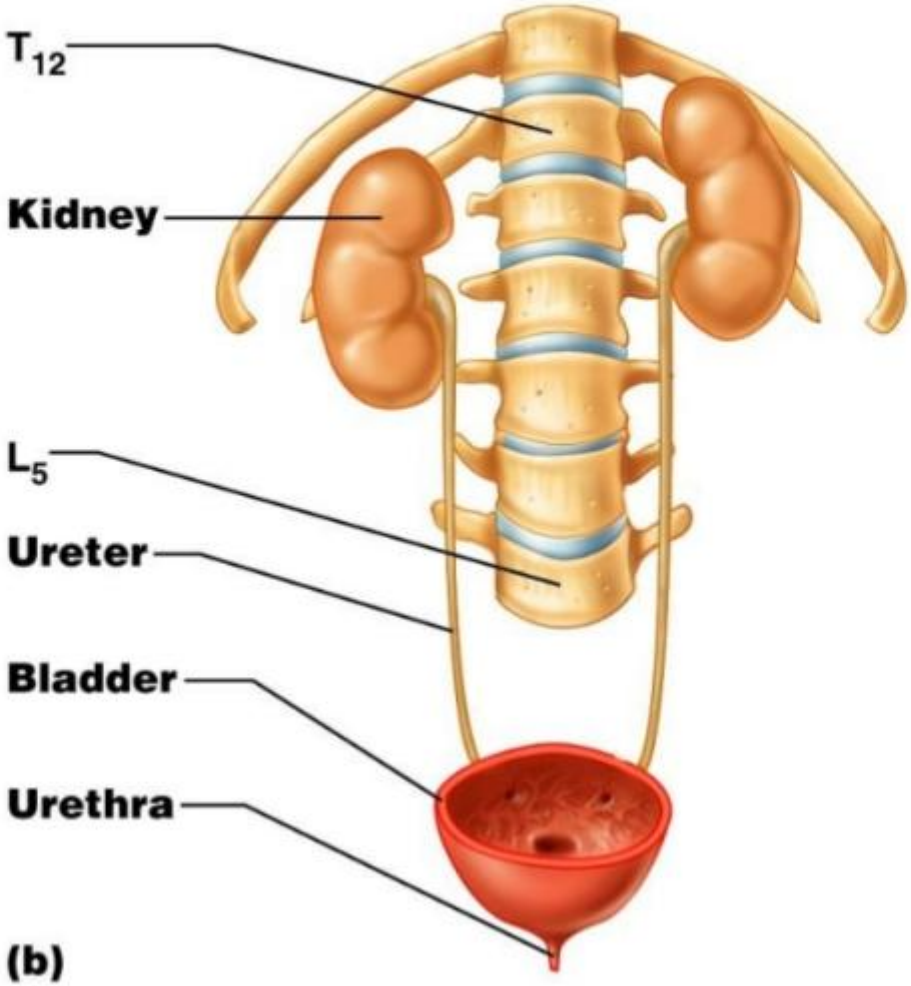
Kidneys

The kidneys are a pair of brownish-red structures located retroperitoneally (behind and outside the peritoneal cavity) on the posterior wall of the abdomen from the 12th thoracic vertebra to the 3rd lumbar vertebra in the adult



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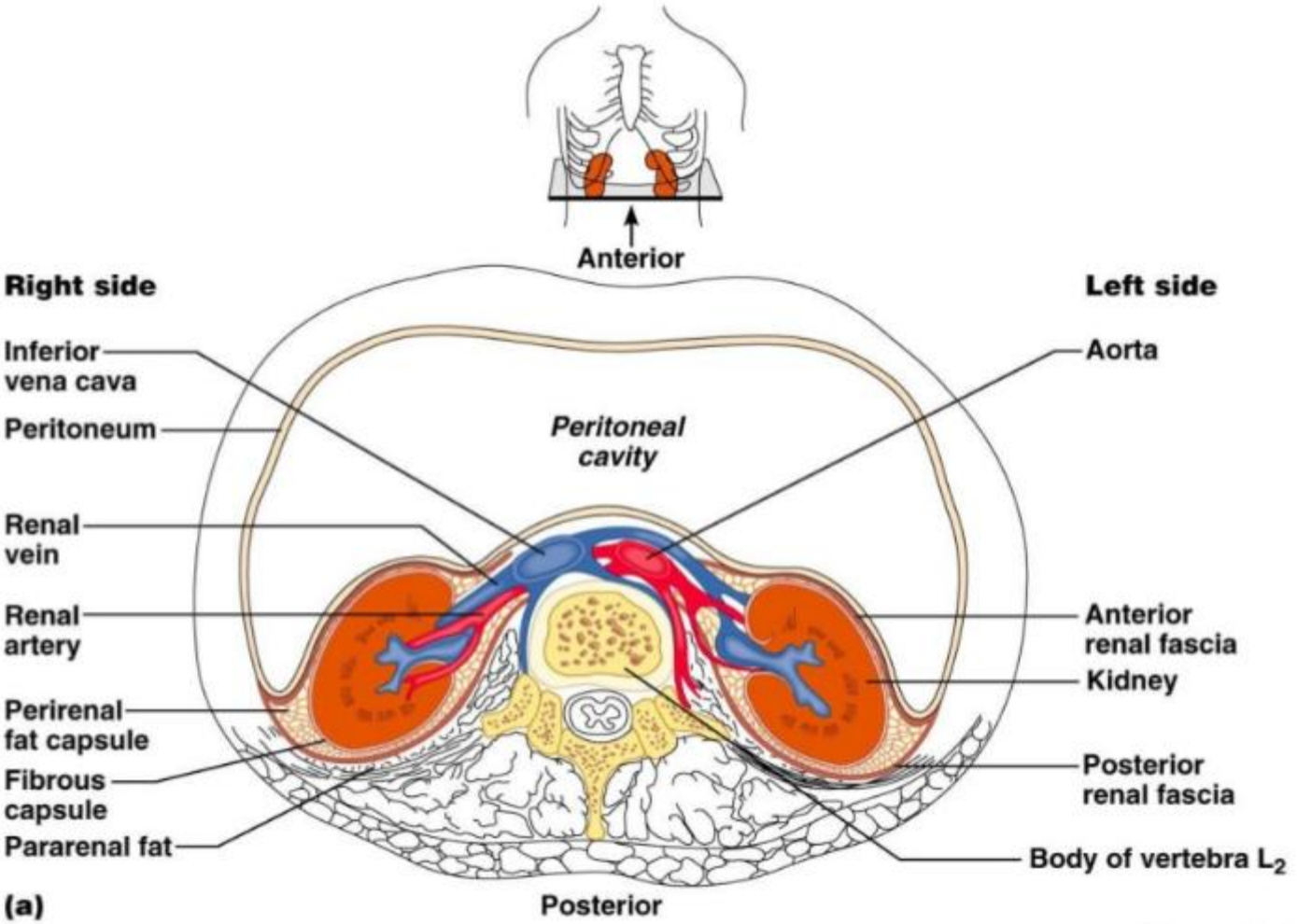
Relationship of the Kidneys to Vertebra and Ribs



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Figure 23.1b

Position of the Kidneys with in the Posterior Abdominal Wall

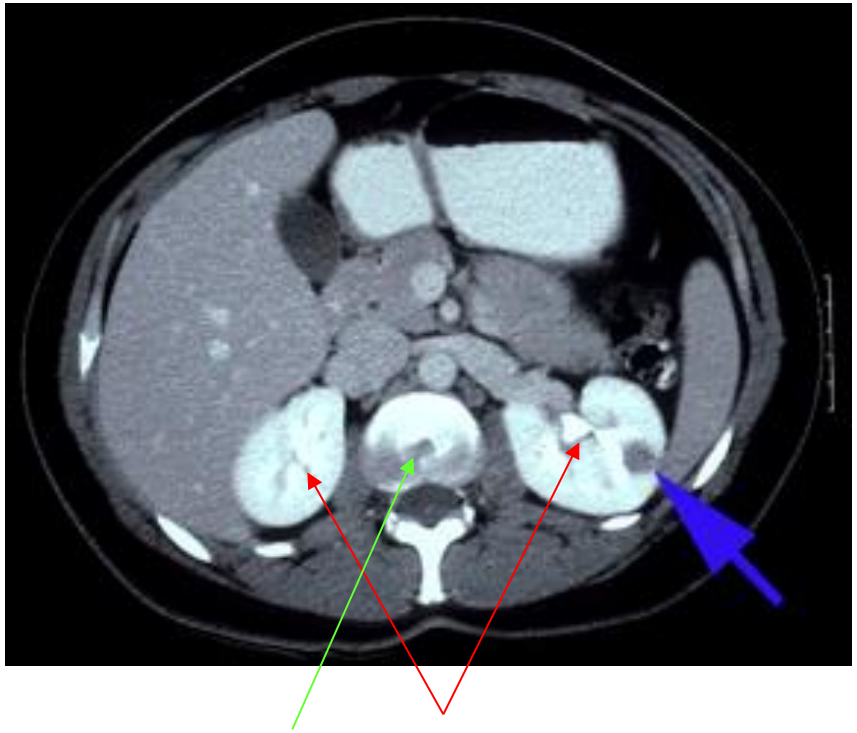


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Figure 23.2a

Position of the Kidneys

CT abdomen with contrast



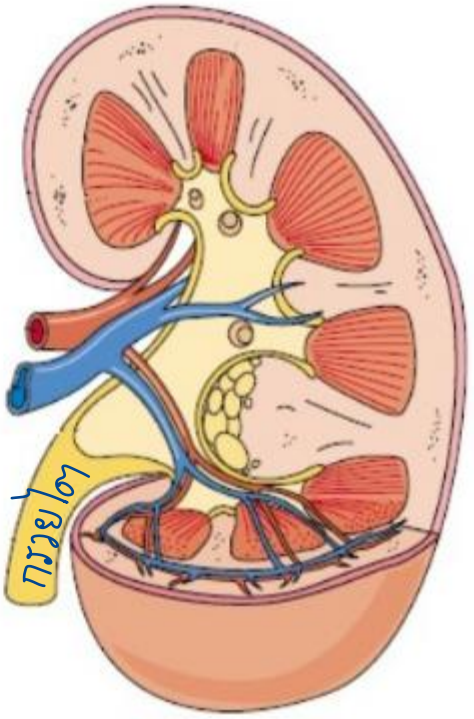
MRI coronal abdomen



Kidneys

- An adult kidney weighs 120 to 170 g (about 4.5 oz) and is 12 (about 4.5 inches) long, 6 cm wide, and 2.5 cm thick.

- The kidneys are well protected by the ribs, muscles, Gerota's fascia, perirenal fat, and the renal capsule, which surround each kidney.

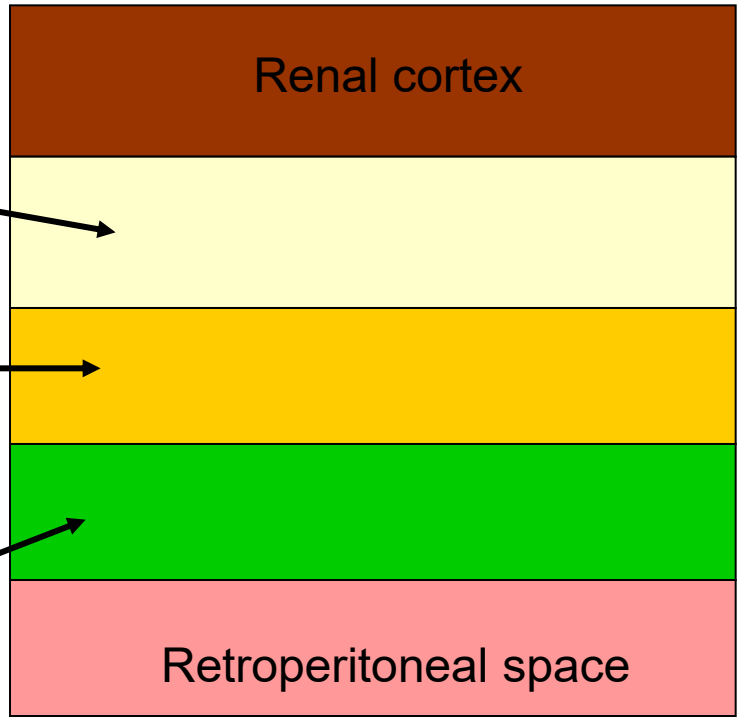


Kidney

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Protection of the Kidneys

- 3 layers of connective tissue:
  - Inner layer- Renal capsule
  - Middle layer- Adipose capsule
  - Outer layer-Renal fascia



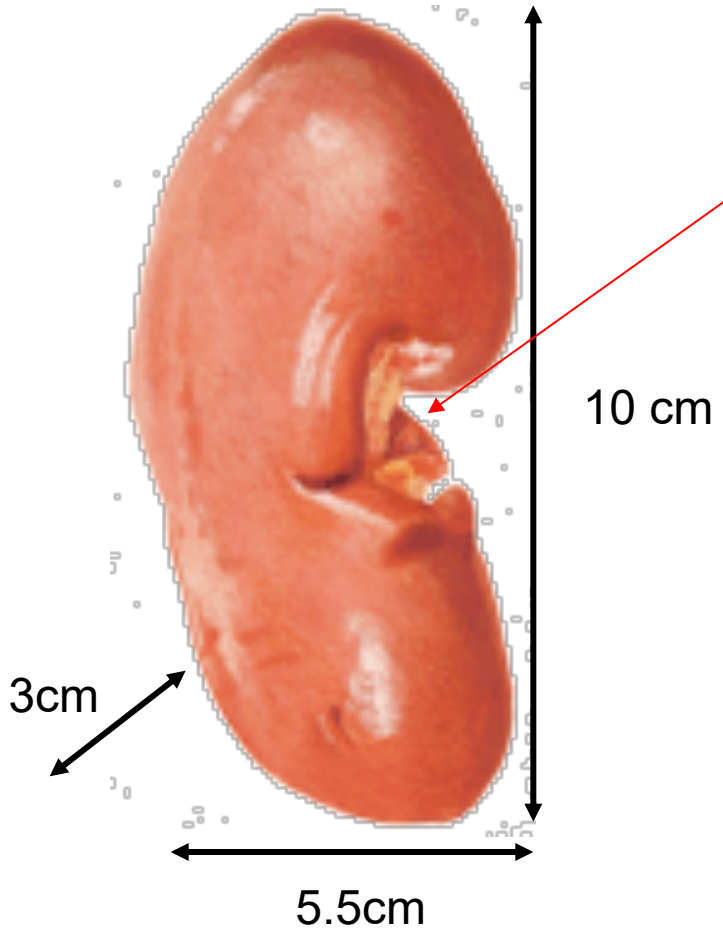
Kidneys

- Urine formation
- Excretion of waste products
- Regulation of electrolytes
- Regulation of acid–base balance
- Control of water balance
- Control of blood pressure
- Renal clearance
- Regulation of red blood cell production
- Synthesis of vitamin D to active form
- Secretion of prostaglandins



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Surface anatomy of the Kidney



- Hilum is located on the medial surface

Kidney Section

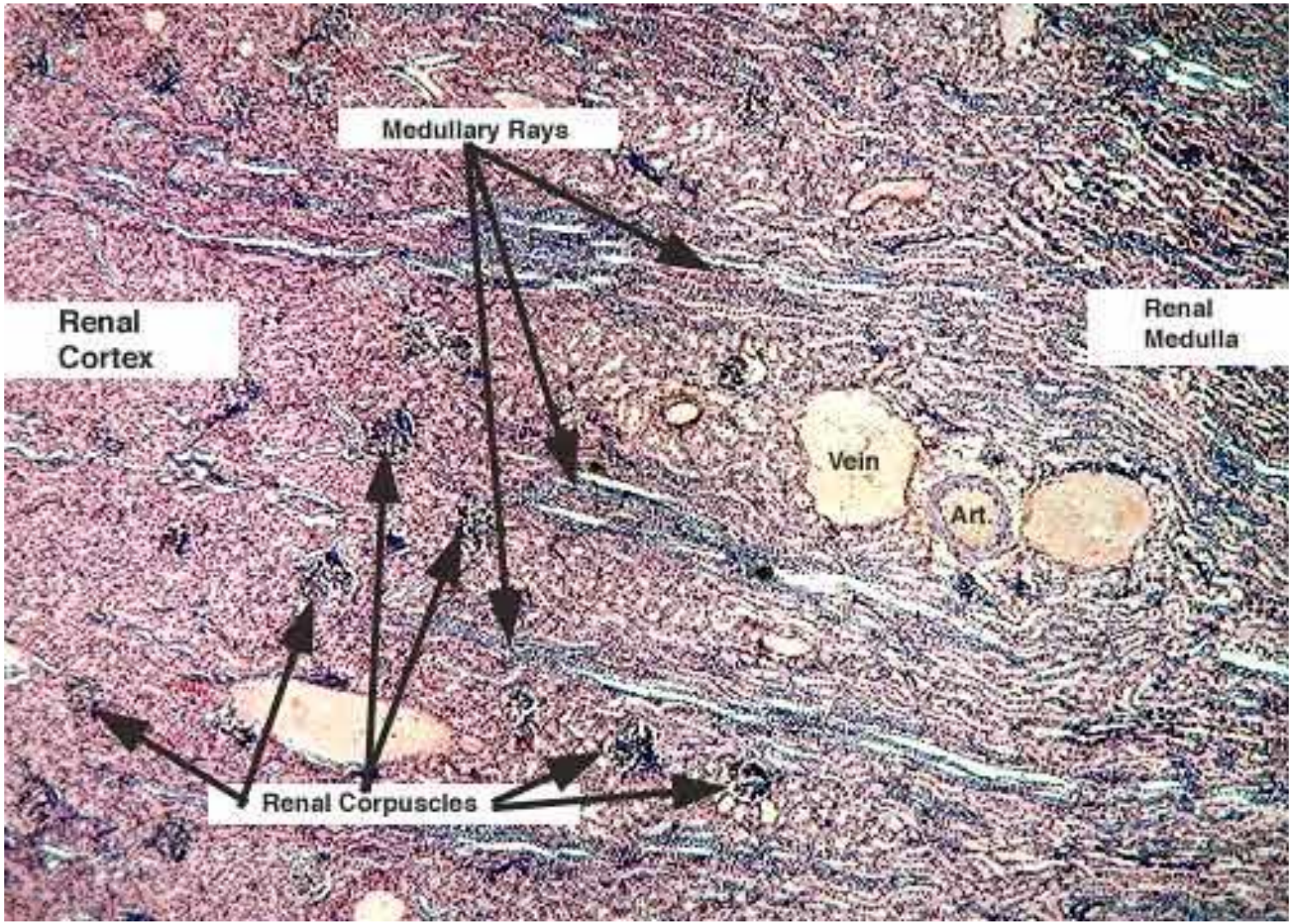


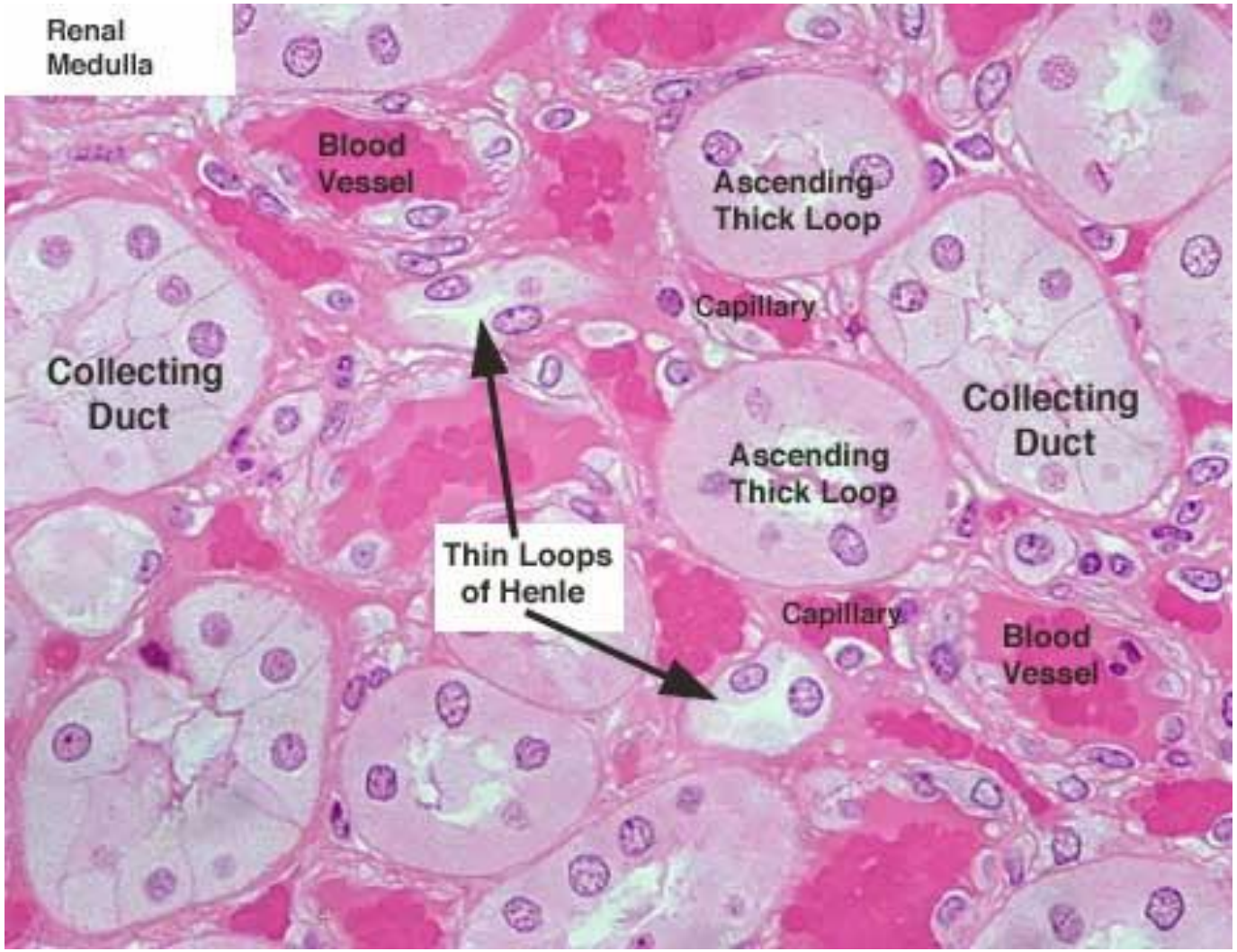
นอก
cortex

สามเหลี่ยมคว่ำ
pyramid
or
Medulla
(เนื้อที่ทอดยาว)

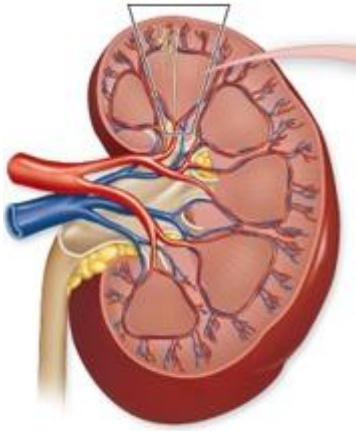
calyx
แบ่งเป็น 2 อัน
Major, minor
Mi ก่อน ค่อย Ma

Renal column
ทางยาว Medulla


Medullary Rays
Renal Cortex
Renal Medulla
Vein
Art.
Renal Corpuscles


Renal Medulla
Blood Vessel
Ascending Thick Loop
Capillary
Collecting Duct
Collecting Duct
Ascending Thick Loop
Thin Loops of Henle
Capillary
Blood Vessel

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ท่อขดส่วนปลาย
Distal convoluted tubule (DCT)
Nephron loop
ท่อขดส่วนต้น
Proximal convoluted tubule
Juxtamedullary nephron
Renal corpuscle โบว์แมน
(PCT)

* จำชาน . ดีๆ)

ที่รวมตัวเรียก Nephron

Proximal convoluted tubule
Renal corpuscle (cut)
Glomerulus
Glomerular capsule
Distal convoluted tubule
Cortical nephron

Cortex
Thick (ขาลงช้วน) descending limb
Thin descending limb (ขาลงผอม)
Thick ascending limb (ขาขึ้นช้วน)
Thin ascending limb (ขาขึ้นผอม)
Nephron loop
Arcuate vessels
Nephron loop
Medulla
→ ลูปออฟเฮนเล
Collecting ducts
ขยะ
ขยะ
Papillary duct
Renal papilla

เข้าไต → กรอง → ขี่
↓
เลือดไป เลือด

ขยะ Medulla

กัรขยะ บ้าน เรา

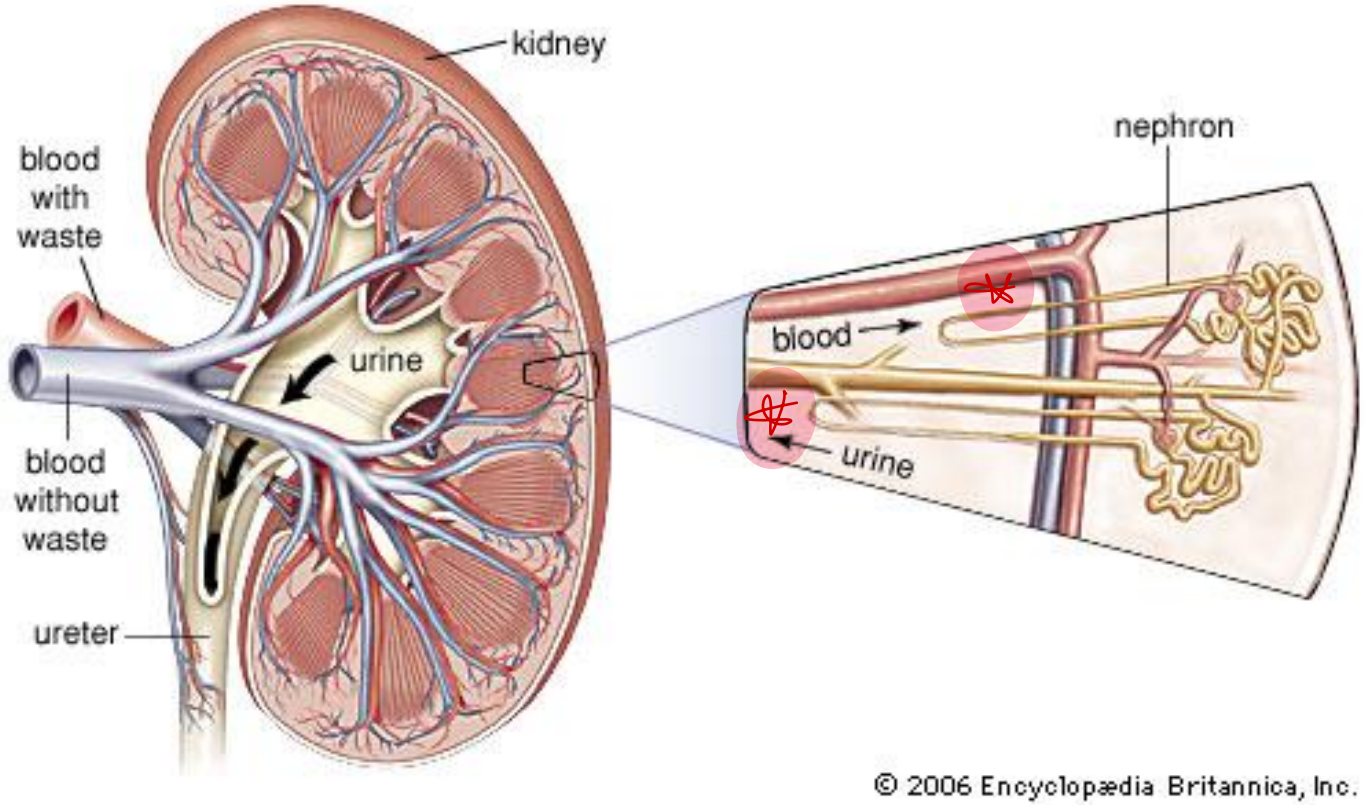
รถขยะ บอรหมู่ (Minor) บ้าน

รถขยะศิทบล (Major)

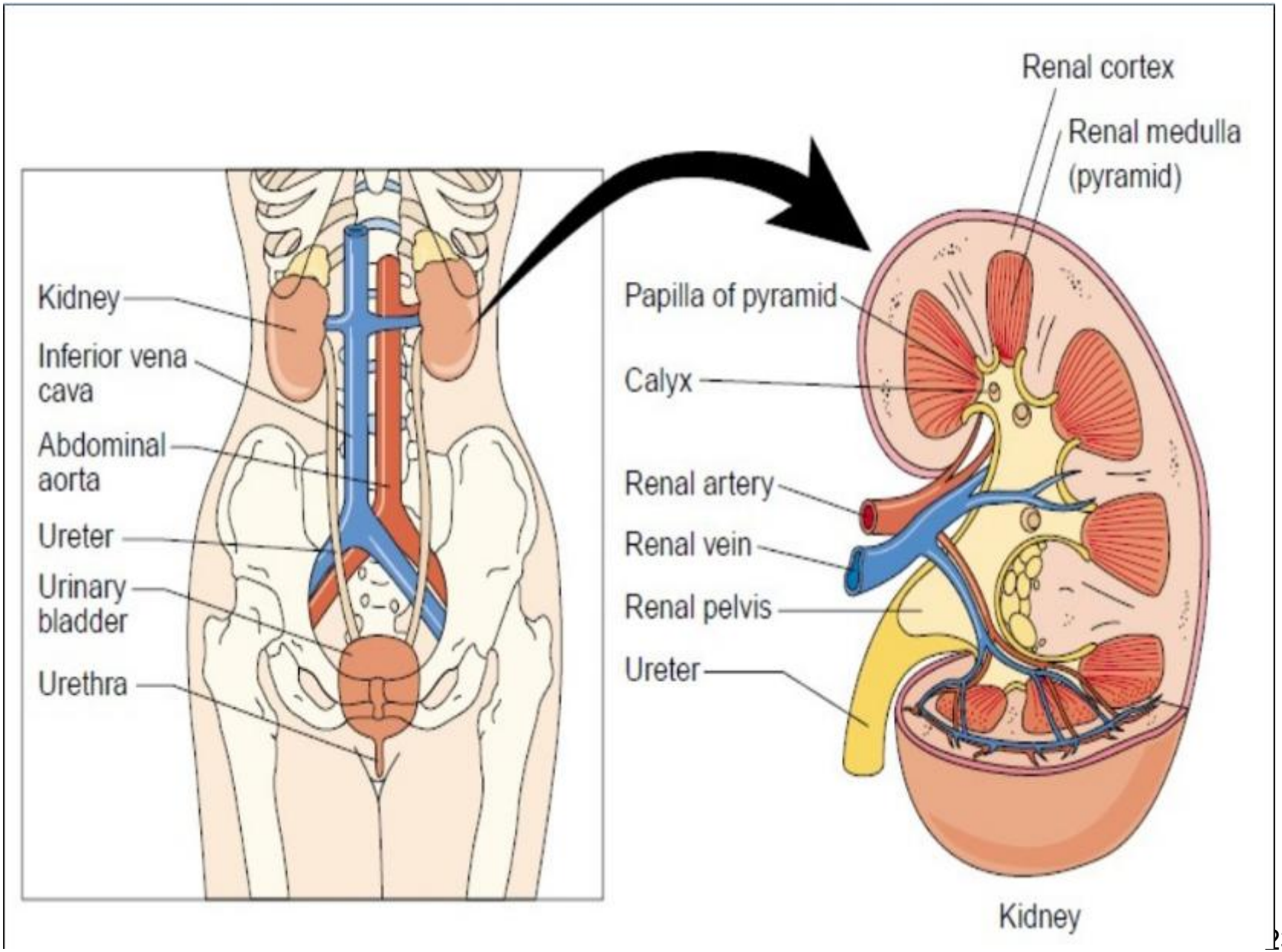
plevis (ส่วนกาง)

Ureter

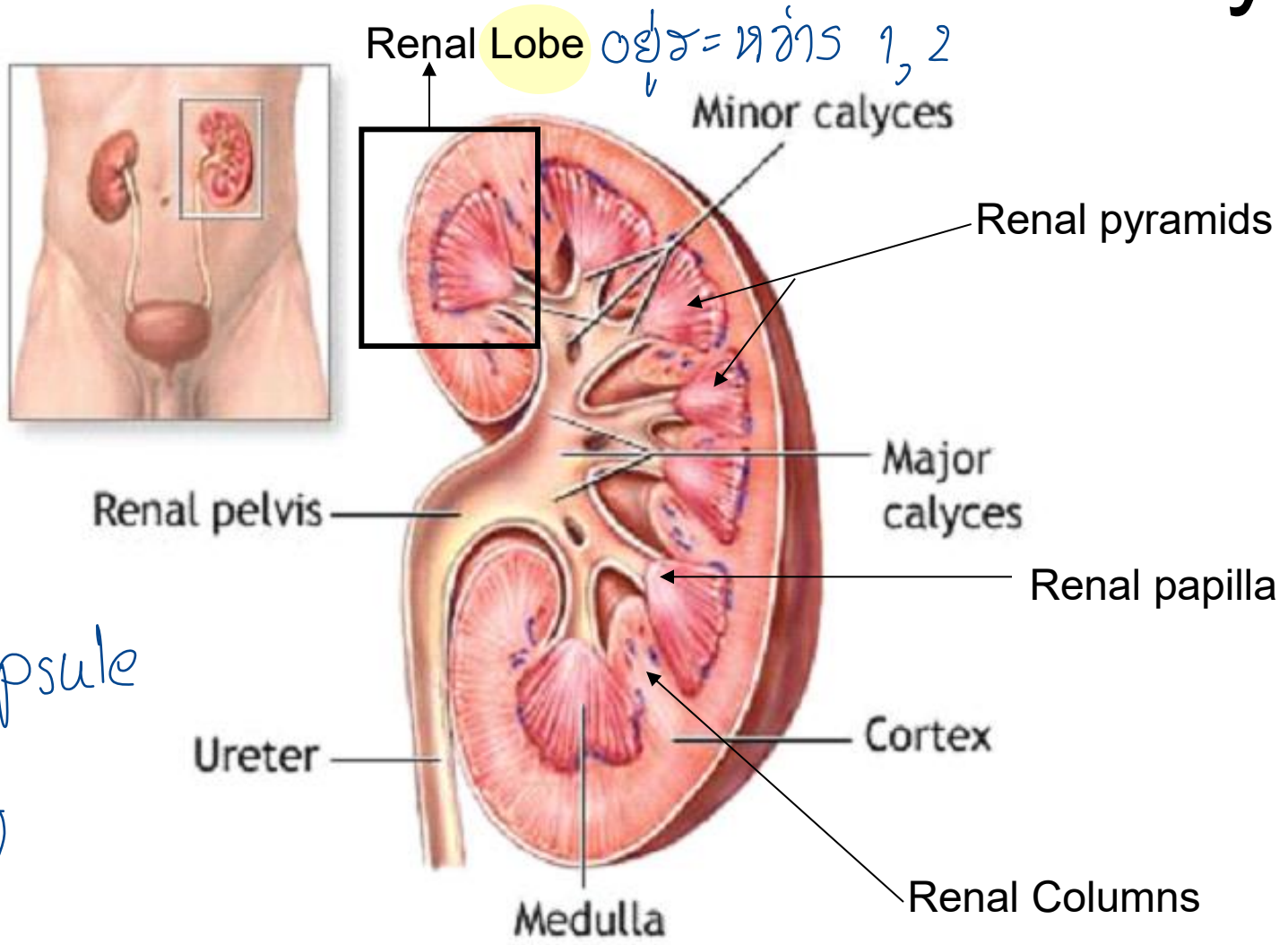
กระเพาะปัส



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Kidney
Inferior vena cava
Abdominal aorta
Ureter
Urinary bladder
Urethra
Renal cortex
Renal medulla (pyramid)
Papilla of pyramid
Calyx
Renal artery
Renal vein
Renal pelvis
Ureter
Kidney

Internal Structure of the Kidney

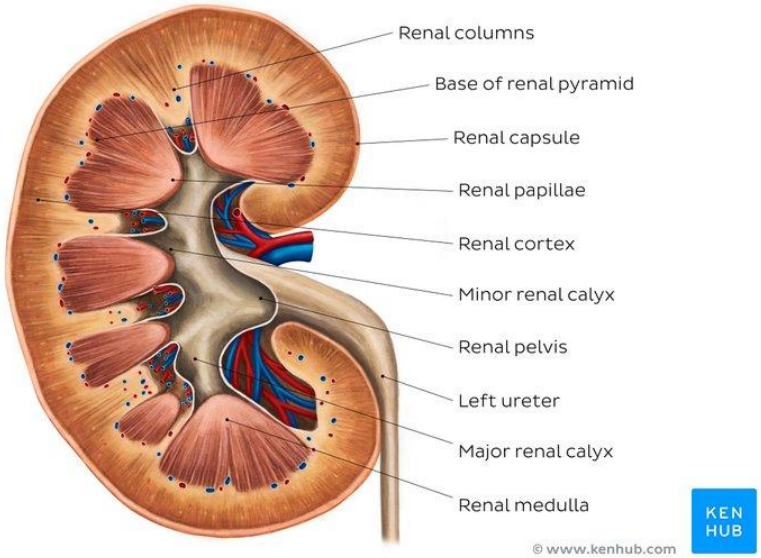


Renal capsule

(หุ้มไต)

Internal Gross Anatomy of the Kidneys

- Frontal section through the kidney
  - Renal cortex
  - Renal pyramids
  - Renal pelvis
    - Major calicies
    - Minor calicies
- Gross vasculature
  - **Renal arteries**
    - Branch into **segmental arteries**



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Summary of Blood Vessels Supplying the Kidney

เข้าไต เข้าโลบไหน โค้ง

Aorta → Renal artery → Segmental artery → Interlobar artery → Arcuate artery → Cortical radiate artery → Afferent arteriole → Glomerulus (capillaries) → Efferent arteriole → Peritubular capillaries and vasa recta → Cortical radiate vein → Arcuate vein → Interlobar vein → Renal vein → Inferior vena cava

แยกไปไหน

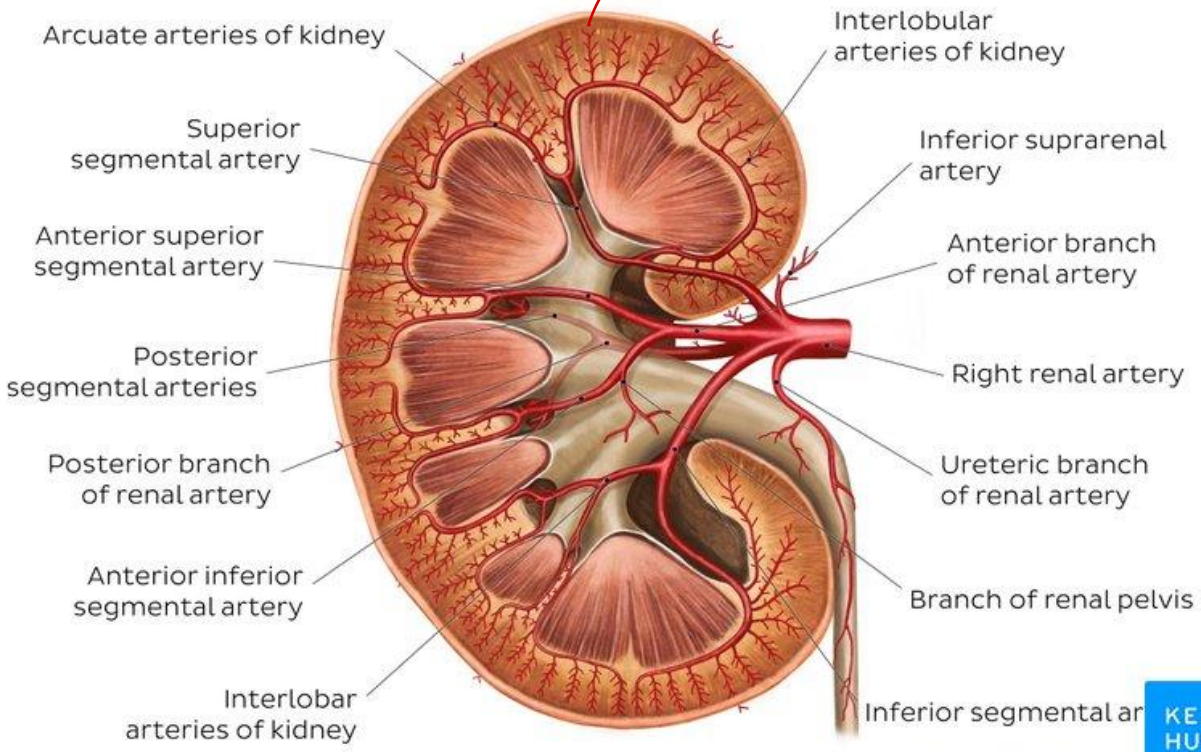
หลอดเลือดแดงเข้า

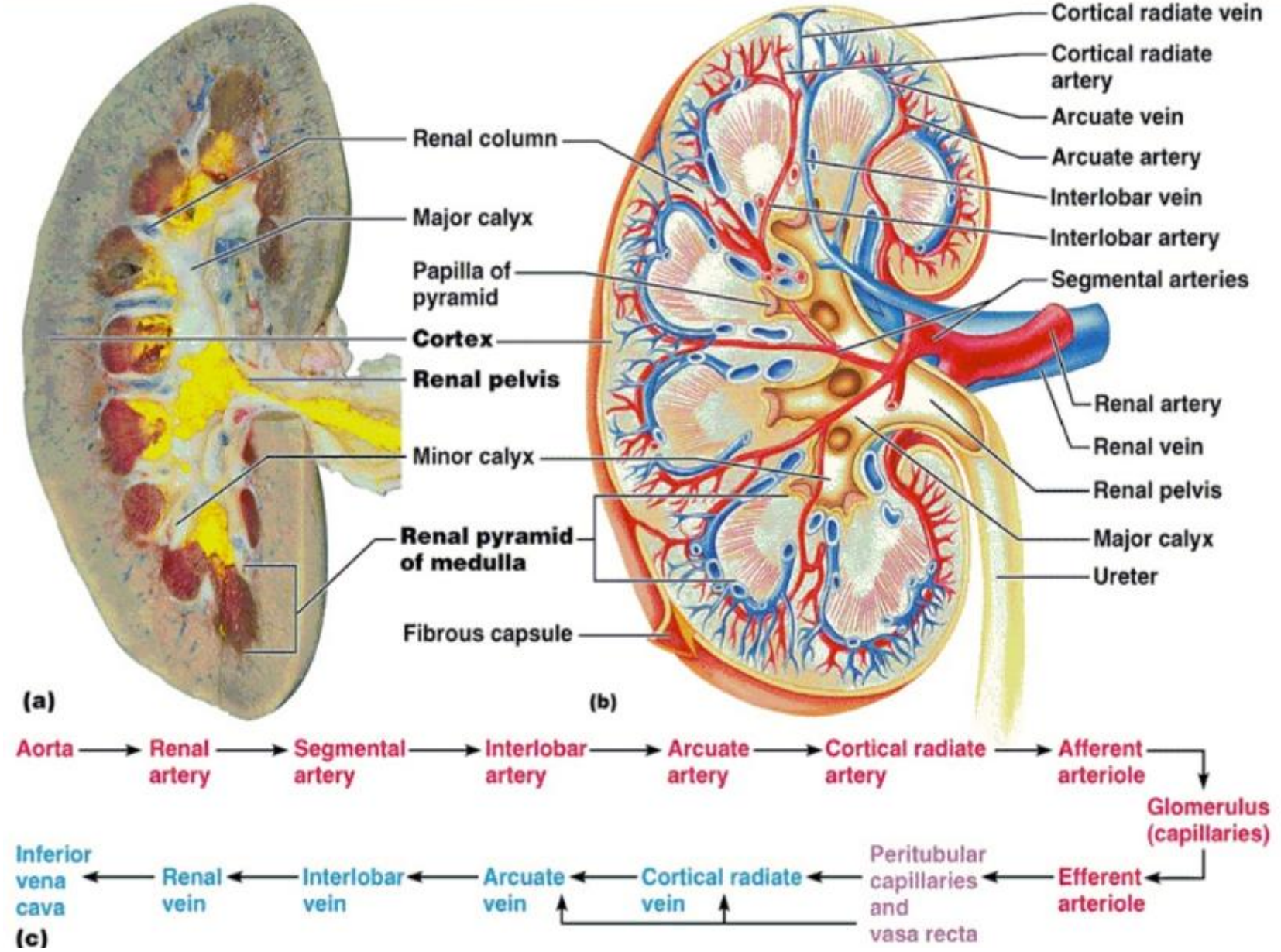
(หุ้มรอบ Glomerulus)

โบว์แมนแคปซูล + Glomerulus (capillaries) = Renal Corpuscle

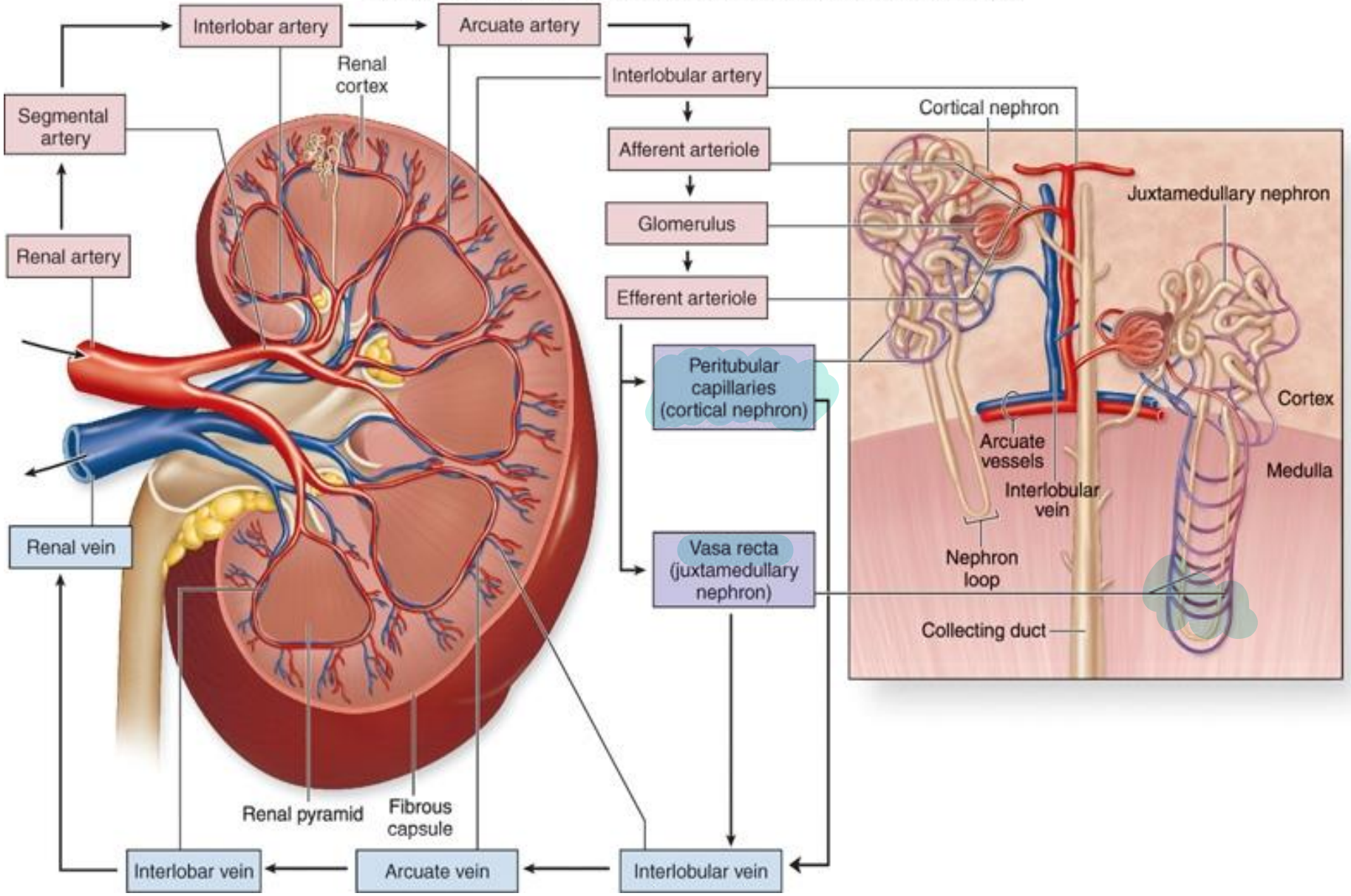
ตลาดนัด

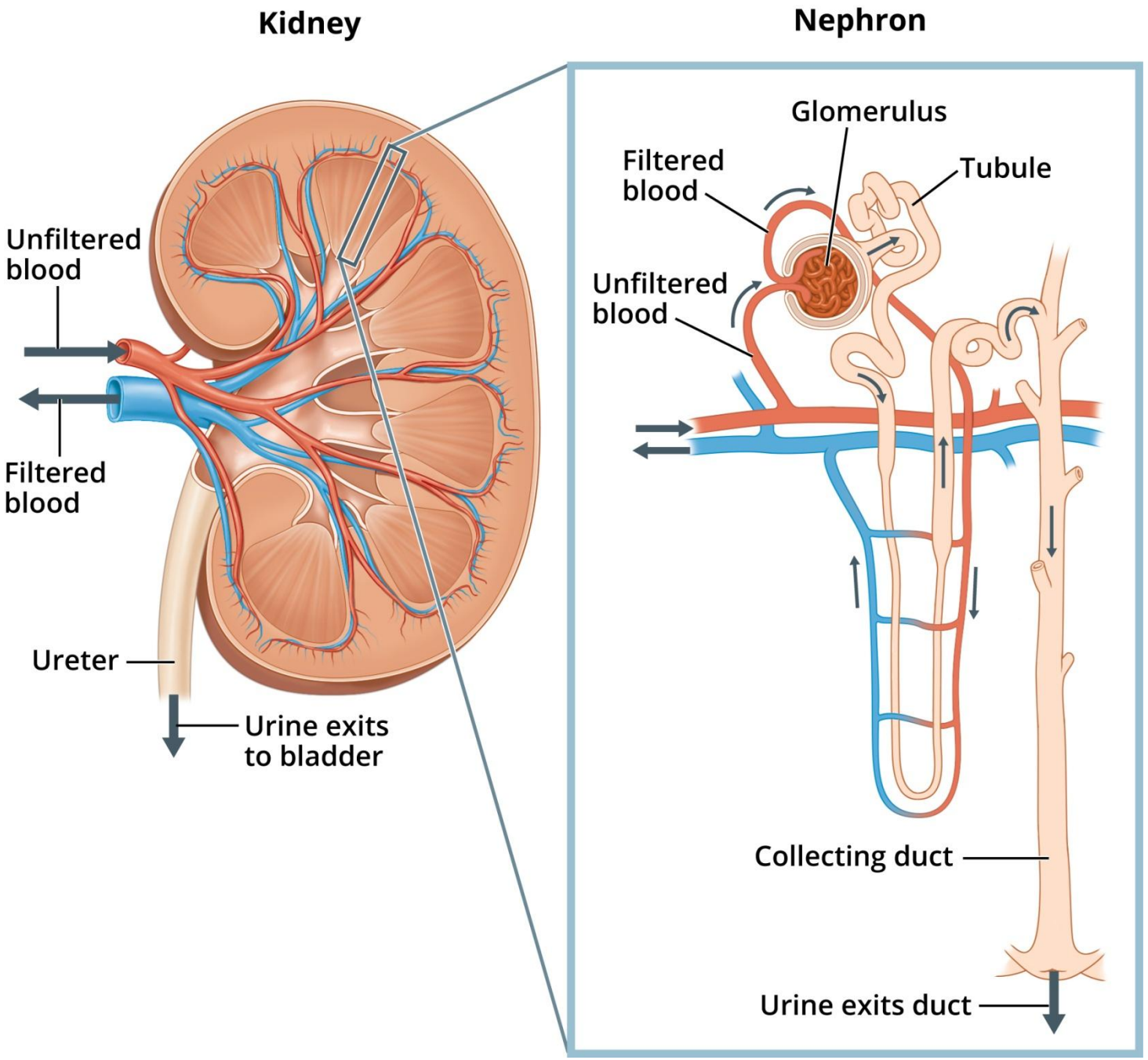
ขาออก



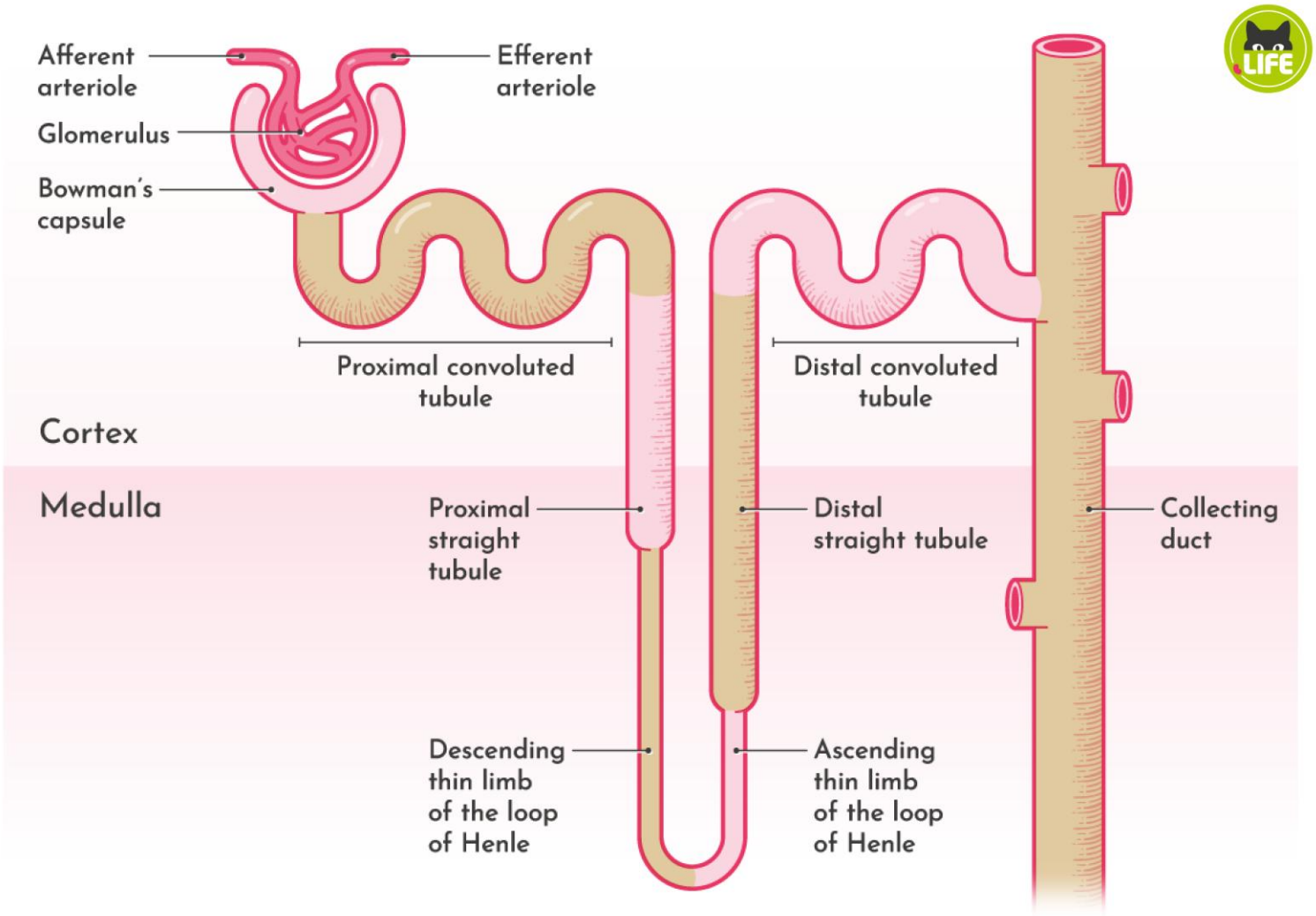

Renal column
Major calyx
Papilla of pyramid
Cortex
Renal pelvis
Minor calyx
Renal pyramid of medulla
Fibrous capsule
(a)
Cortical radiate vein
Cortical radiate artery
Arcuate vein
Arcuate artery
Interlobar vein
Interlobar artery
Segmental arteries
Renal artery
Renal vein
Renal pelvis
Major calyx
Ureter
(b)
Aorta → Renal artery → Segmental artery → Interlobar artery → Arcuate artery → Cortical radiate artery → Afferent arteriole → Glomerulus (capillaries) → Efferent arteriole → Peritubular capillaries and vasa recta → Cortical radiate vein → Arcuate vein → Interlobar vein → Renal vein → Inferior vena cava
(c)

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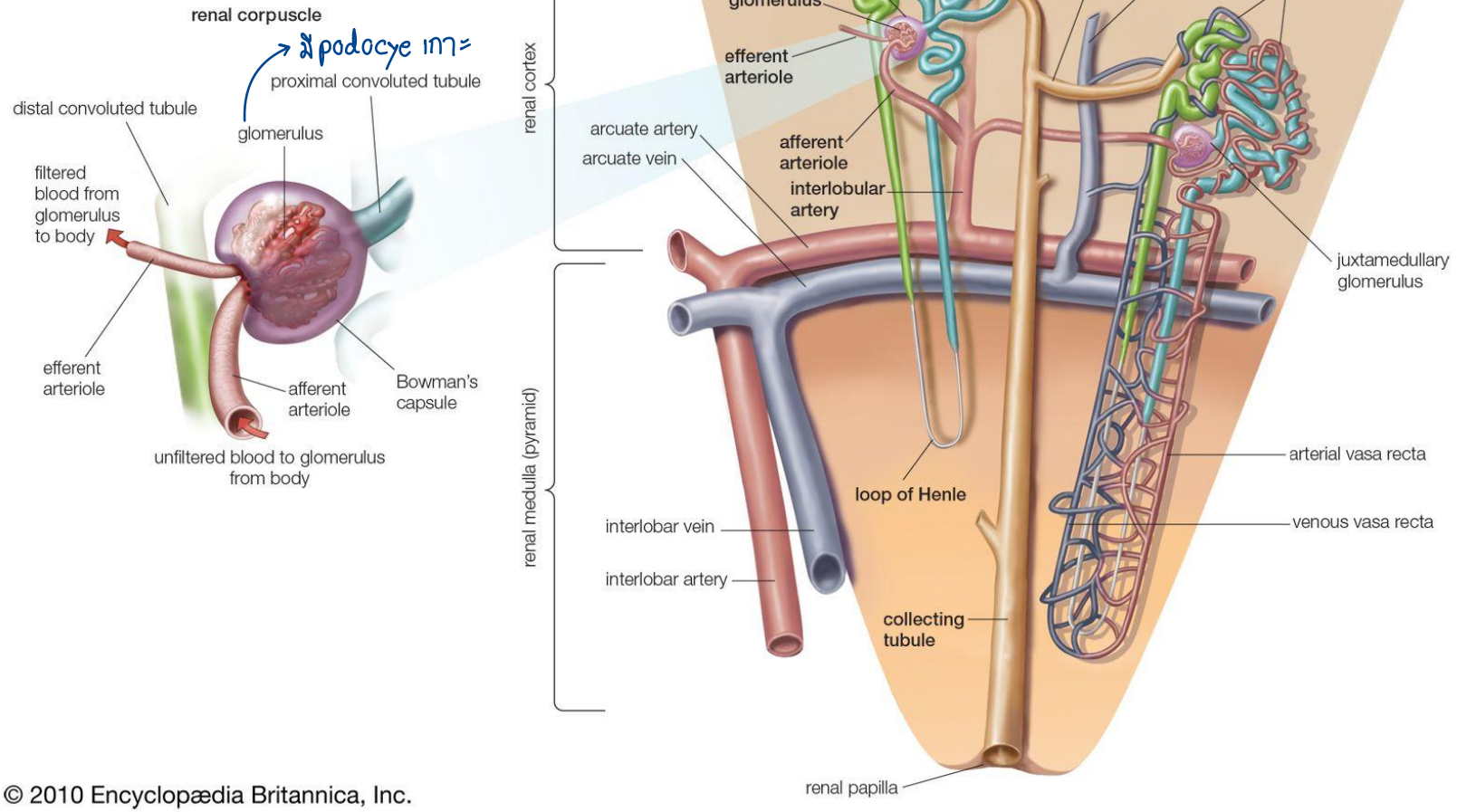



Kidney
Nephron
Unfiltered blood
Filtered blood
Ureter
Urine exits to bladder
Glomerulus
Filtered blood
Tubule
Unfiltered blood
Collecting duct
Urine exits duct

Nephron

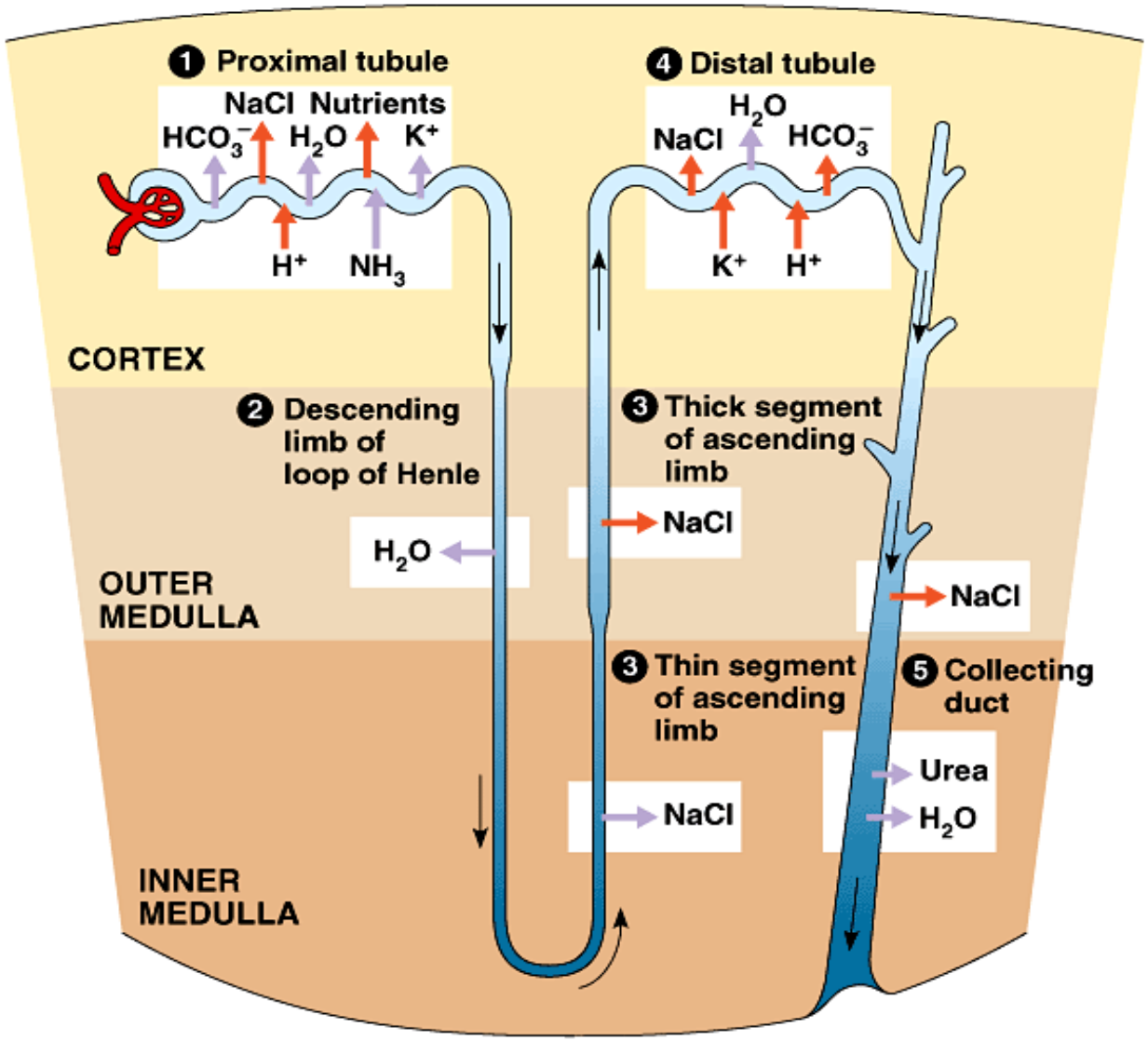

Afferent arteriole
Efferent arteriole
Glomerulus
Bowman's capsule
Proximal convoluted tubule
Distal convoluted tubule
Cortex
Medulla
Proximal straight tubule
Distal straight tubule
Collecting duct
Descending thin limb of the loop of Henle
Ascending thin limb of the loop of Henle

Nephron



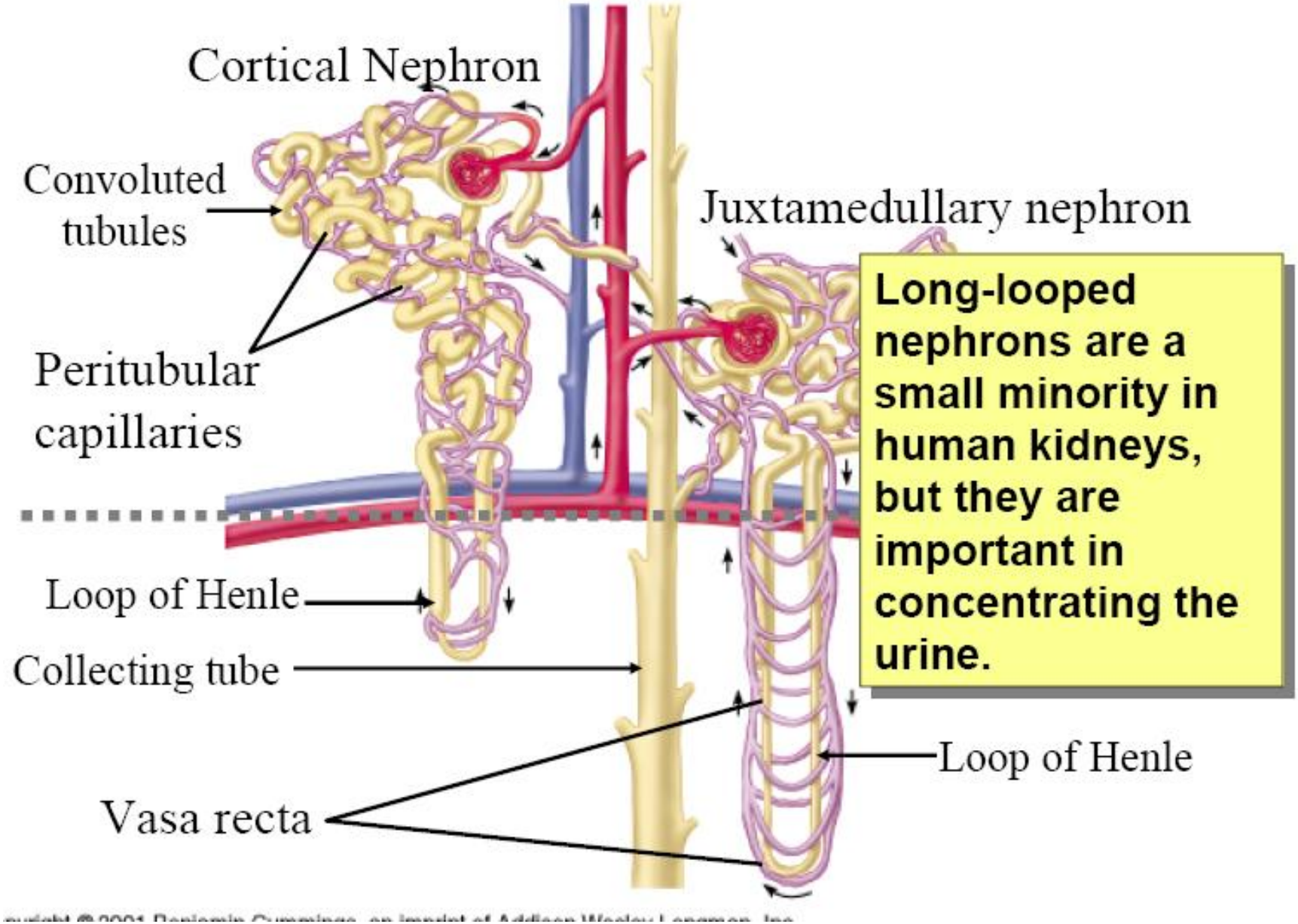
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Nephron-Tubular System



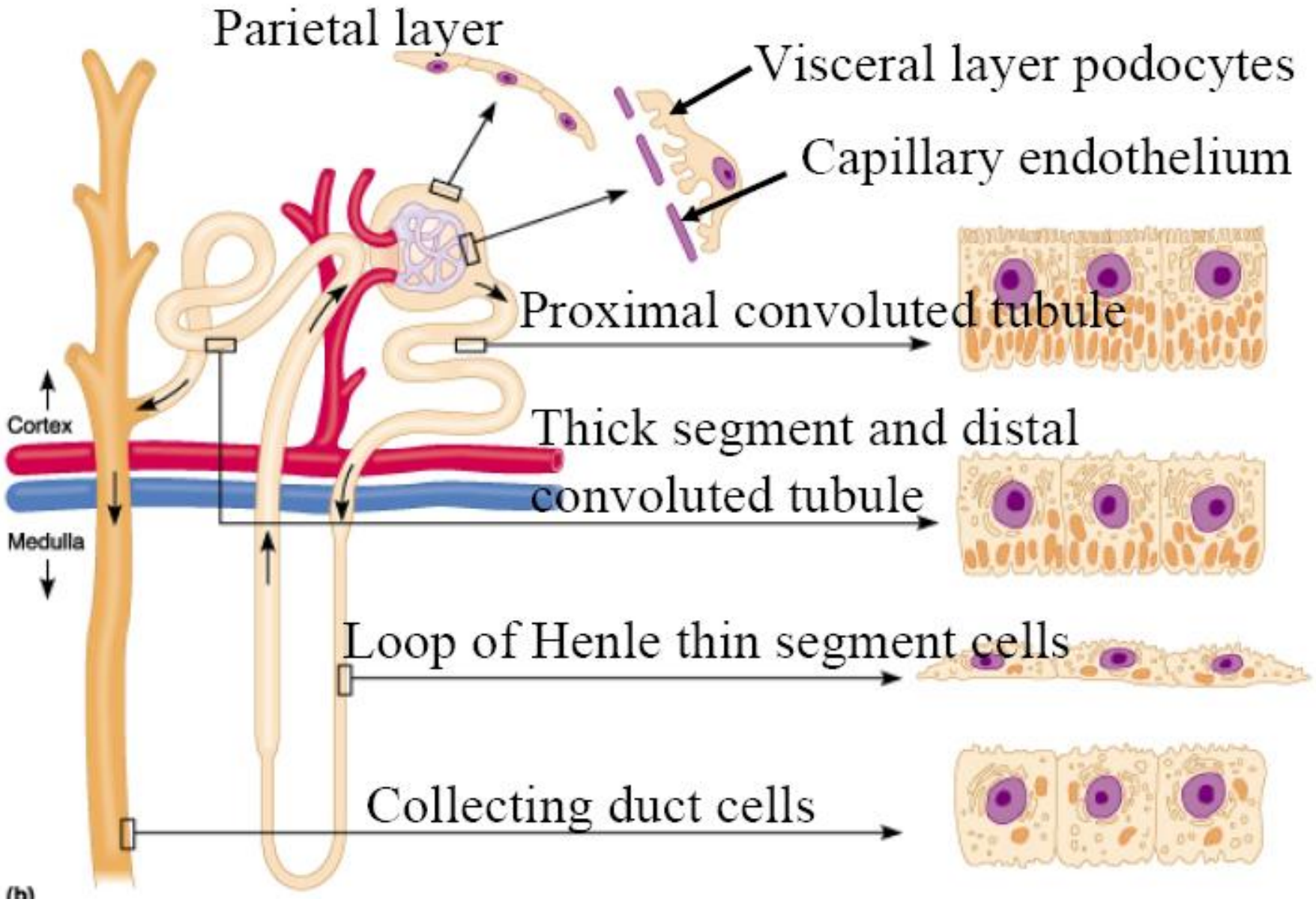
1. Proximal convoluted tubule
2. Descending loop of Henle
3. Ascending loop of Henle
4. Distal convoluted tubule
5. Collecting duct

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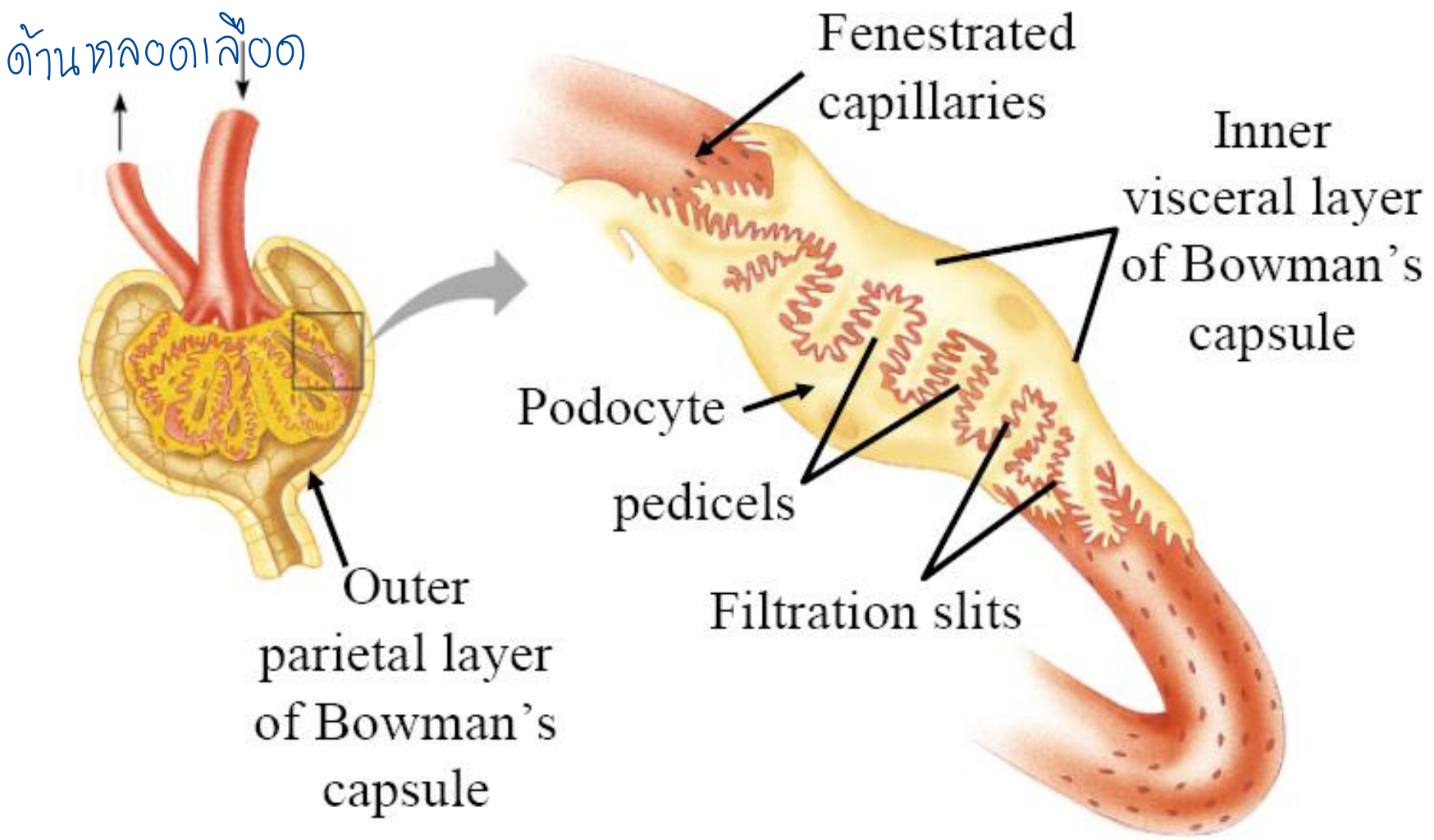


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Histology of the Nephron

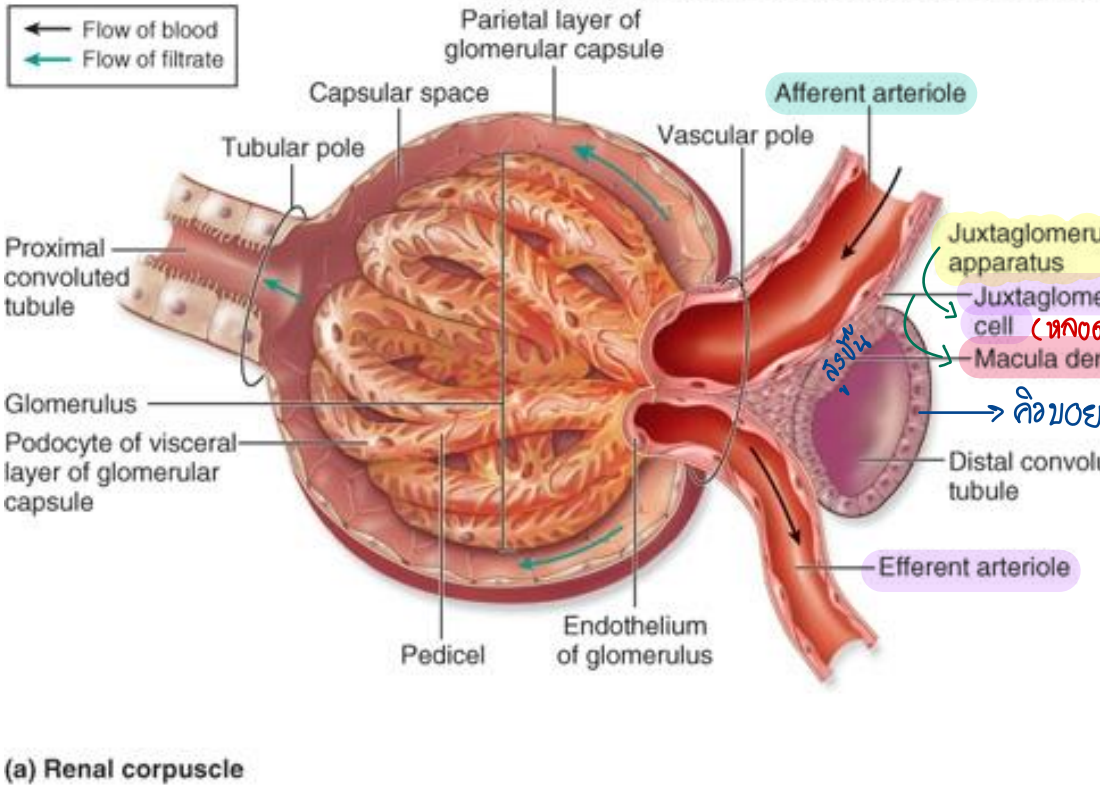


The Glomerulus and Bowman's Capsule



ตาจสีท่อมาอยู่ใกล้หลอดเลือด

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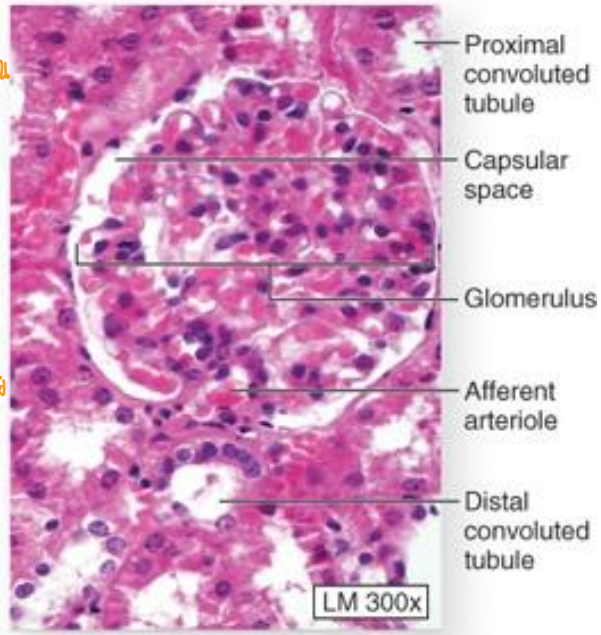
(a) Renal corpuscle

ฮอร์โมนเรนิน (ก.ดัน)

(หลอดเลือด)

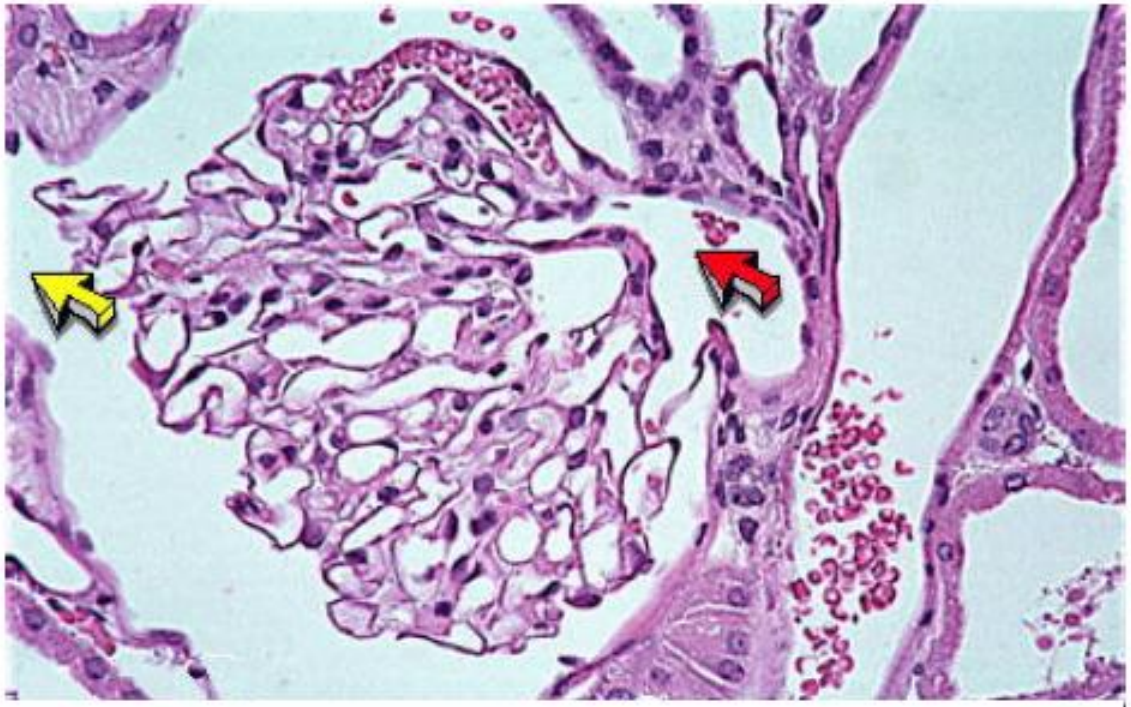
(ท่อ)

คุม Na ของร่างกาย



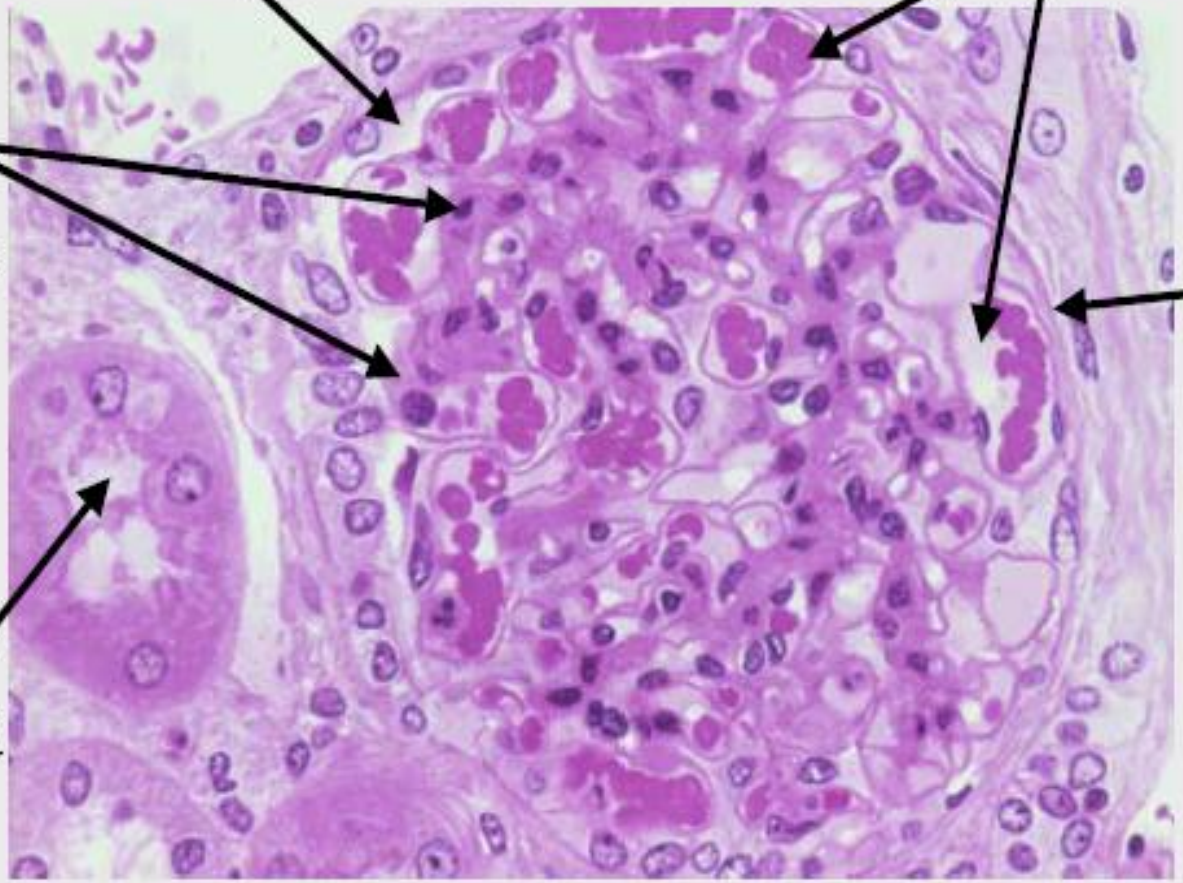
(b)

Glomerular Structures

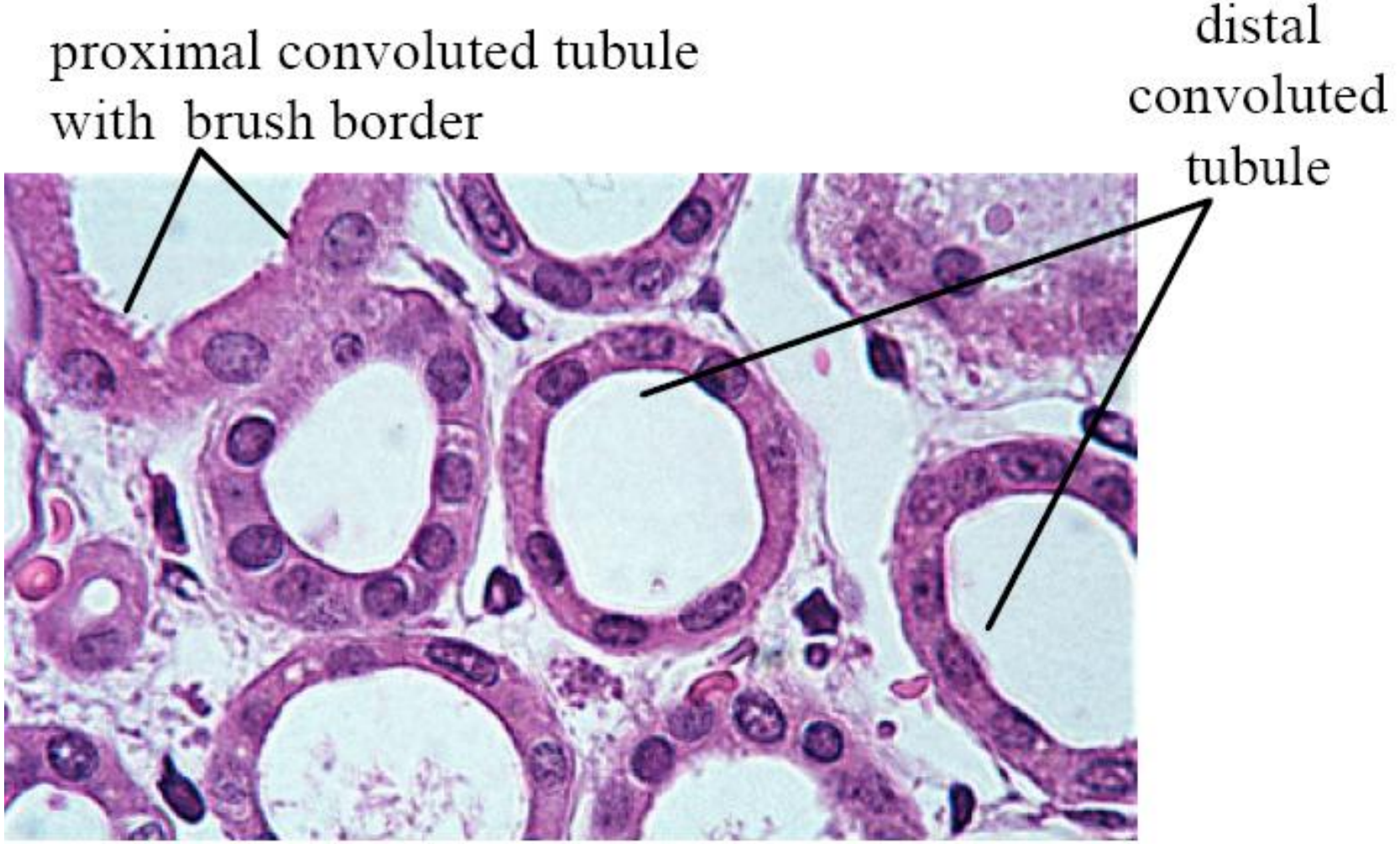


Red indicates the afferent arteriole where it enters the glomerulus at the **vascular pole**. Yellow arrow indicates the opening of the Bowman's capsule into the proximal convoluted tubule.

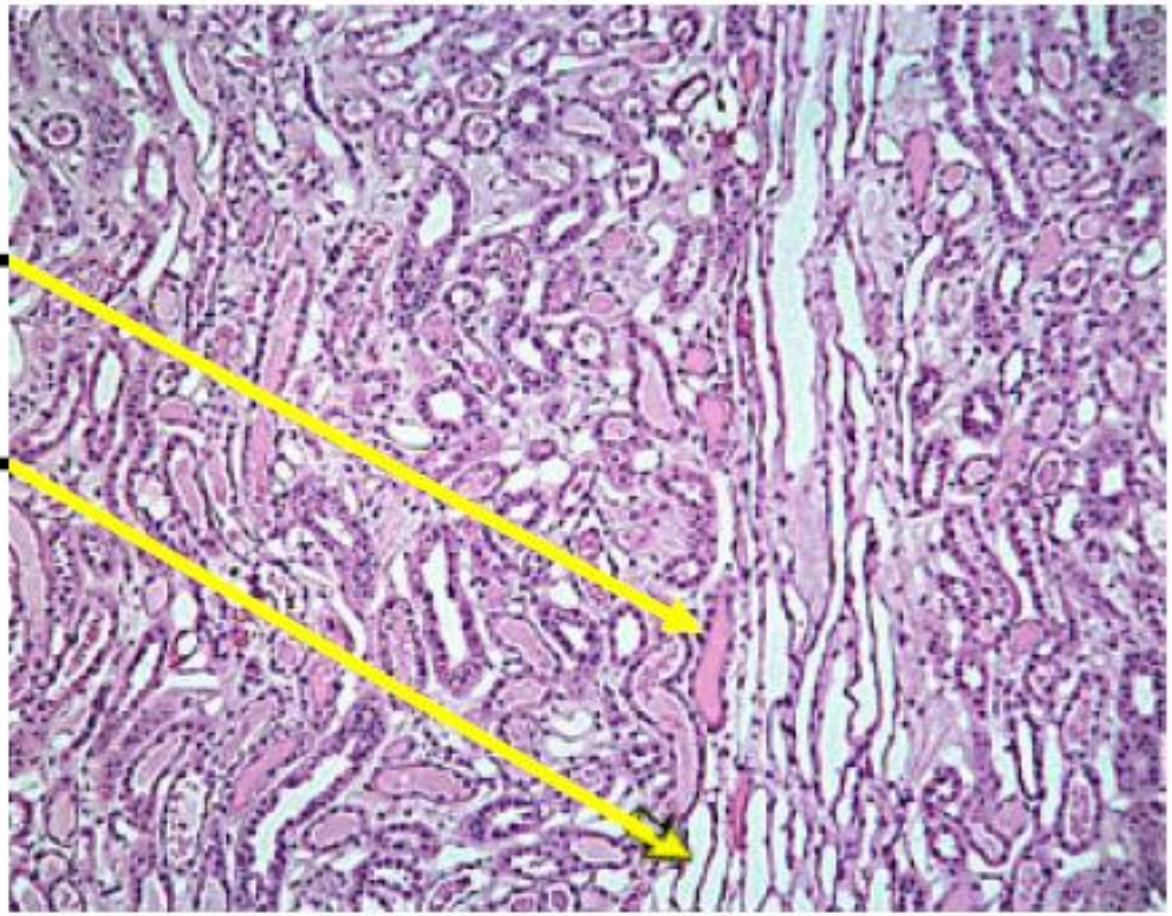
Detailed Glomerular Structure



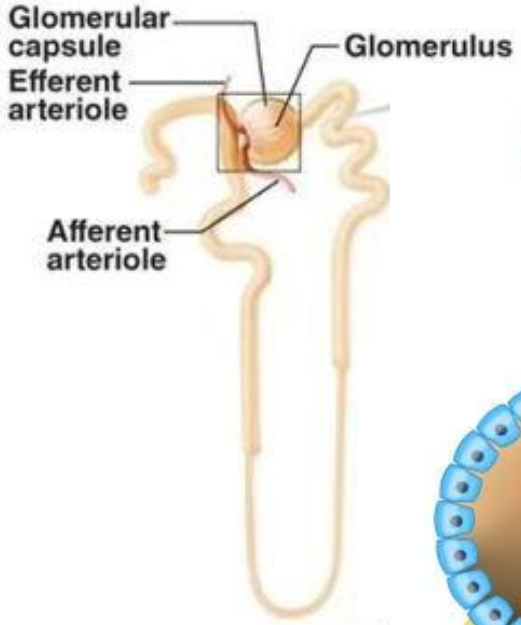
Convoluted Tubules



Renal Medulla



Juxtaglomerular Apparatus (JGA)



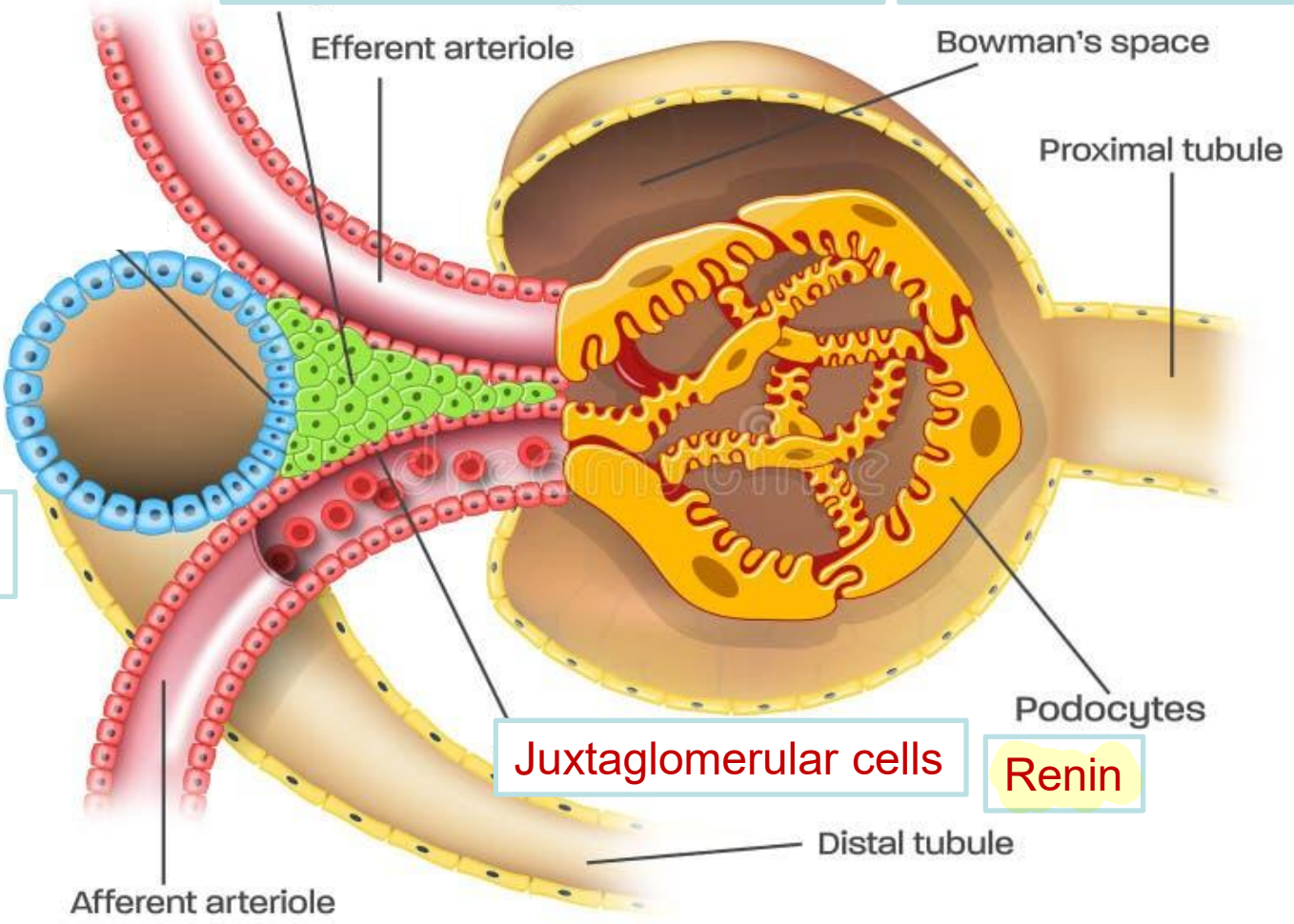
เก็บกินเชื้อโรค

Extraglomerular mesangial cells

ค้ำจุน Phagocytosis

Macula densa

โซเดียม

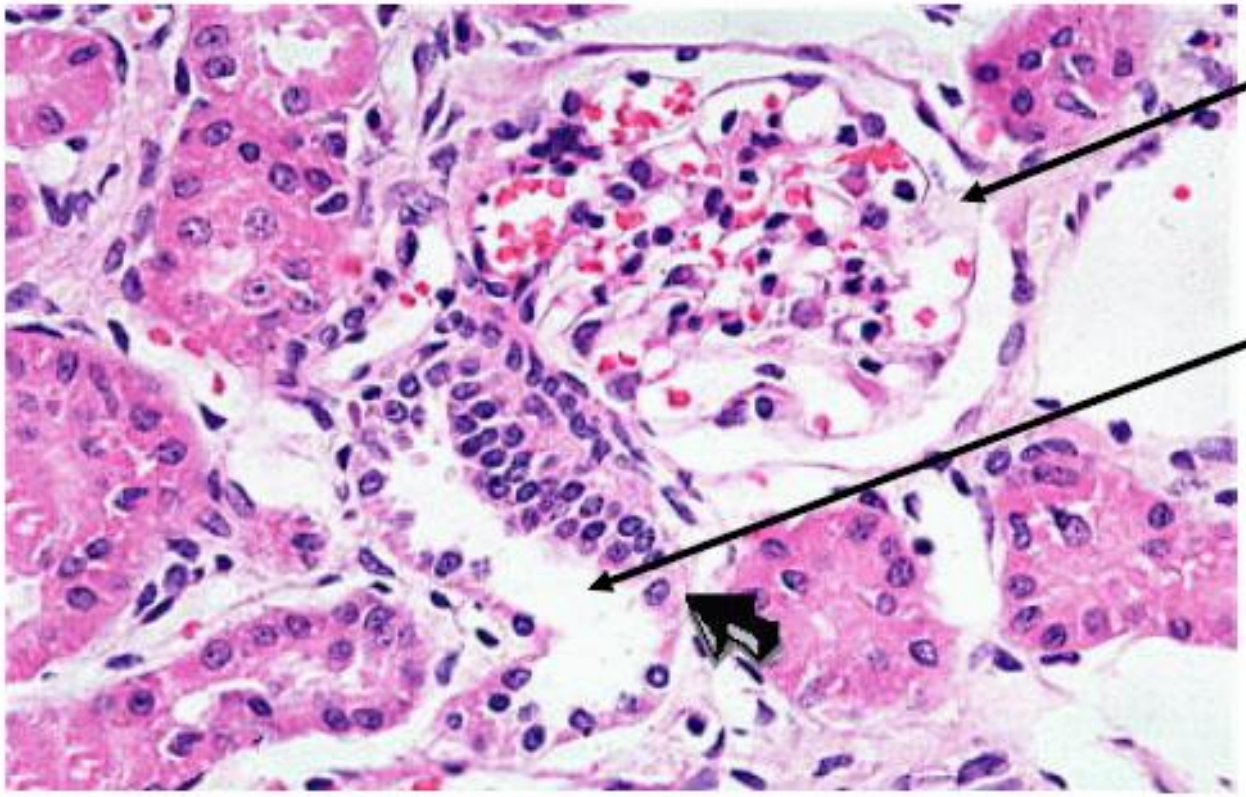


Functions of the JGA Cells:

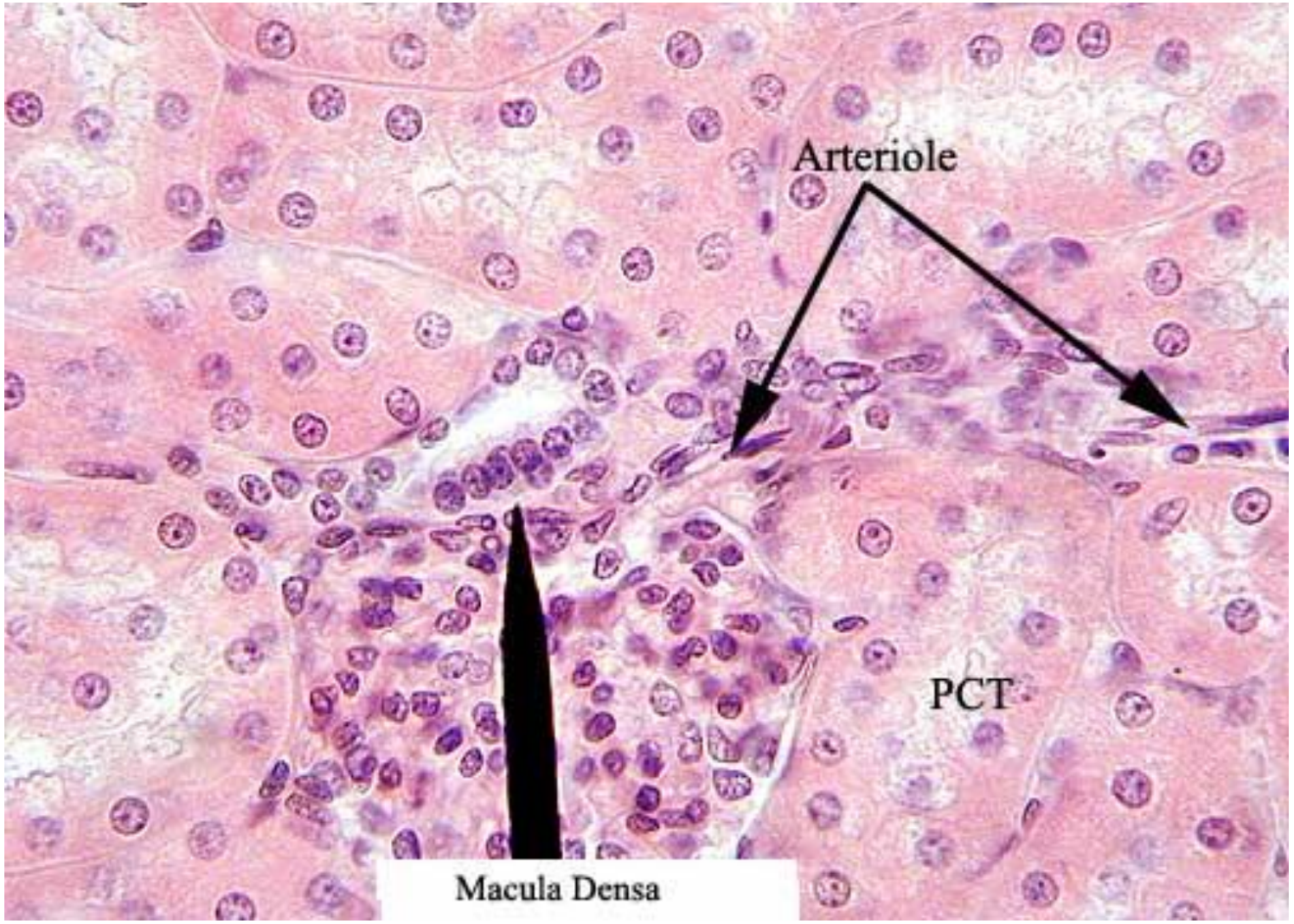
Macula densa cells sense the glomerular filtration rate via the salt (Na^+) concentration in the distal tubule.

Juxtaglomerular cells secrete renin into the blood of the arterioles. They are modified smooth muscle cells which can also vasoconstrict or vasodilate.

The Macula Densa



The macula densa (black) monitors the distal tubular fluid salt concentration so that the juxtaglomerular apparatus can control the filtration rate (via tubuloglomerular feedback).


Arteriole
PCT
Macula Densa

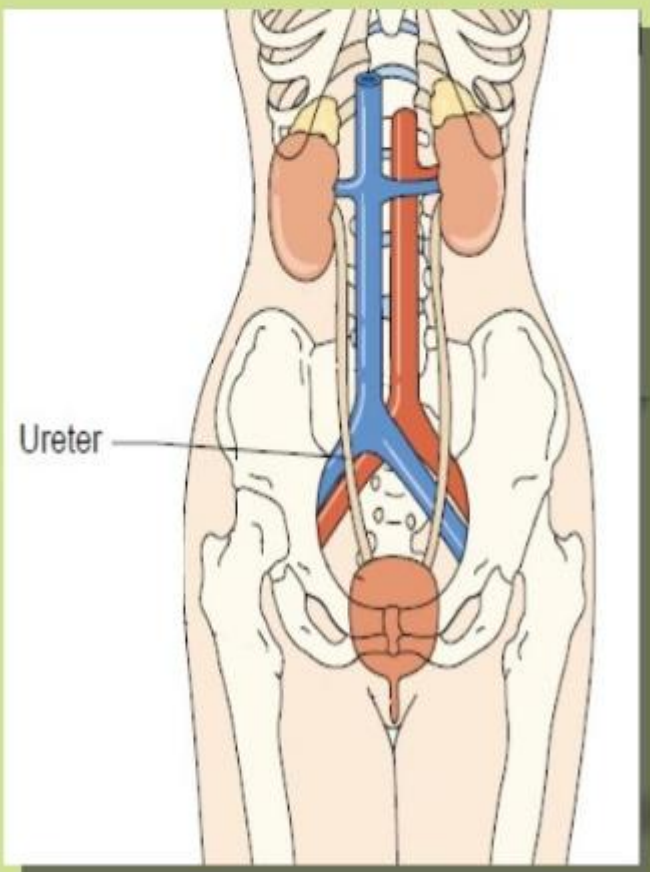
The Formation of Urine

- 3 processes involved in the formation of urine.
  - Simple filtration
  - Selective reabsorbtion
    - Hormonal control-
      - » Parathyroid hormone, calcitonin
      - » Anti diuretic hormone
      - » Aldosterone
  - Secretion

Ureters

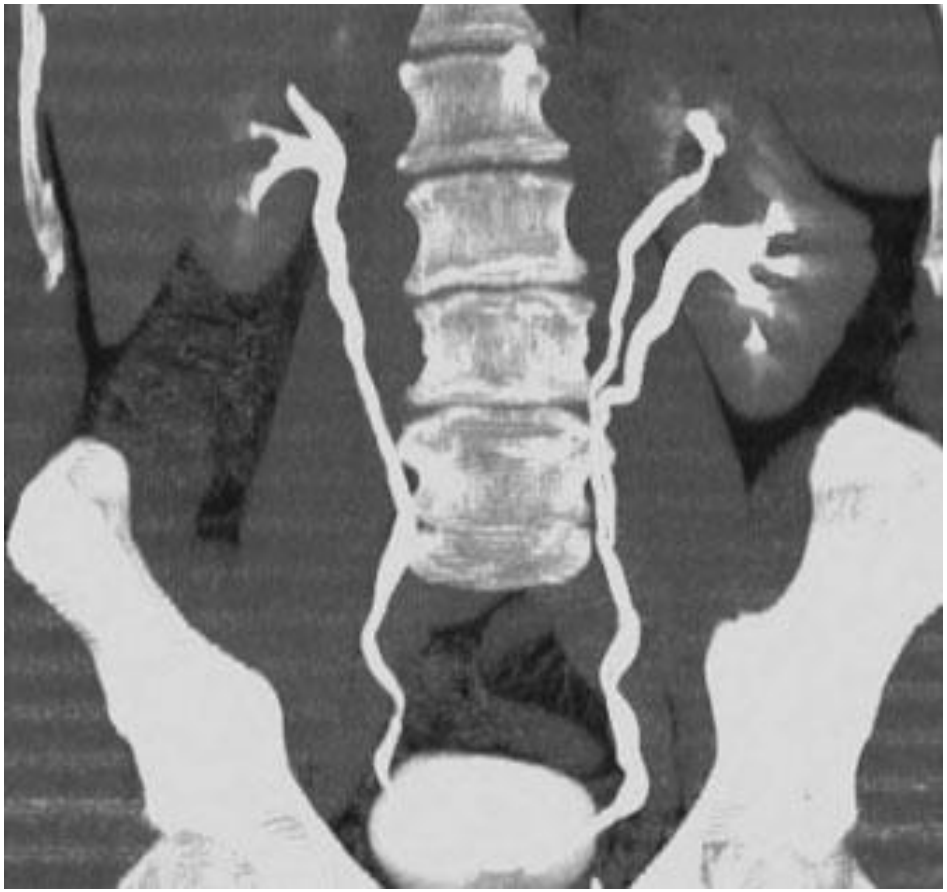
Urine, which is formed within the nephrons, flows into the ureter, a long fibromuscular tube that connects each kidney to the bladder.

The ureters are narrow, muscular tubes, each 24 to 30 cm long, that originate at the lower portion of the renal pelvis and terminate in the trigone of the bladder wall.



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Ureters



- Superiorly บน
  - Continuous with the renal pelvis
- Inferiorly ล่าง
  - Pass through the abdominal cavity, behind the peritoneum, infront of the psoas muscle, into the pelvic cavity where they enter the posterior wall of the bladder
- 25-30 cm in length

Ureters

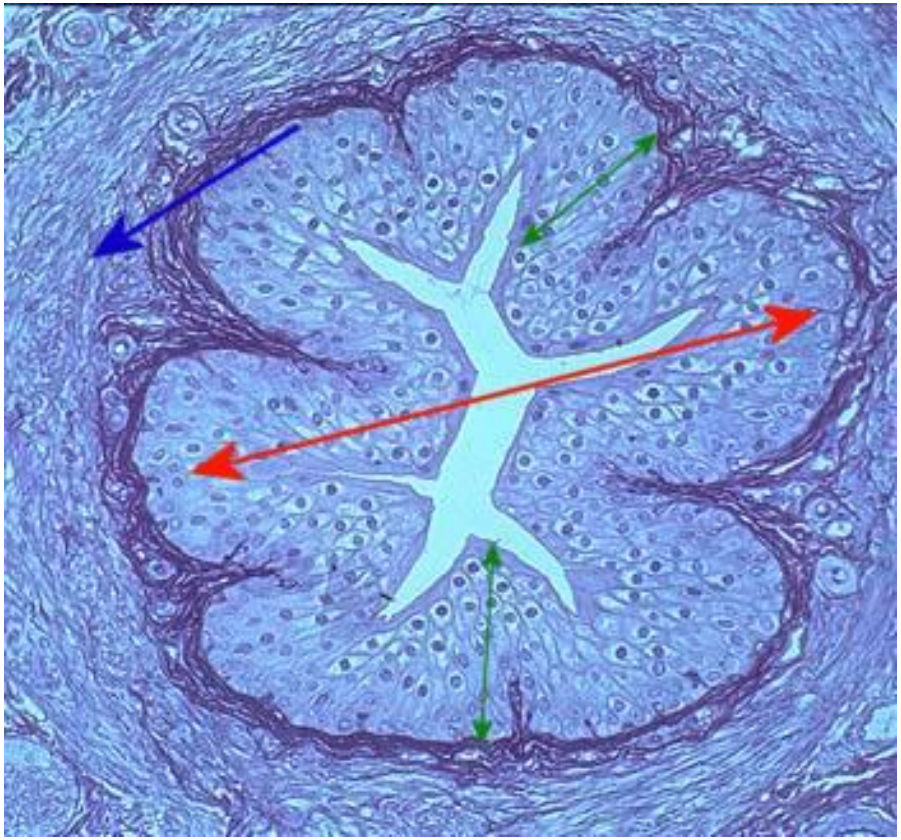
There are three narrowed areas of each ureter:

- ureteropelvic junction
- ureteral segment
- ureterovesical junction



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Ureter- Cross Section



- 3 layers of tissue
  - Outer layer
    - Fibrous tissue
  - Middle layer
    - Muscle
  - Inner layer
    - Epithelium

Ureters

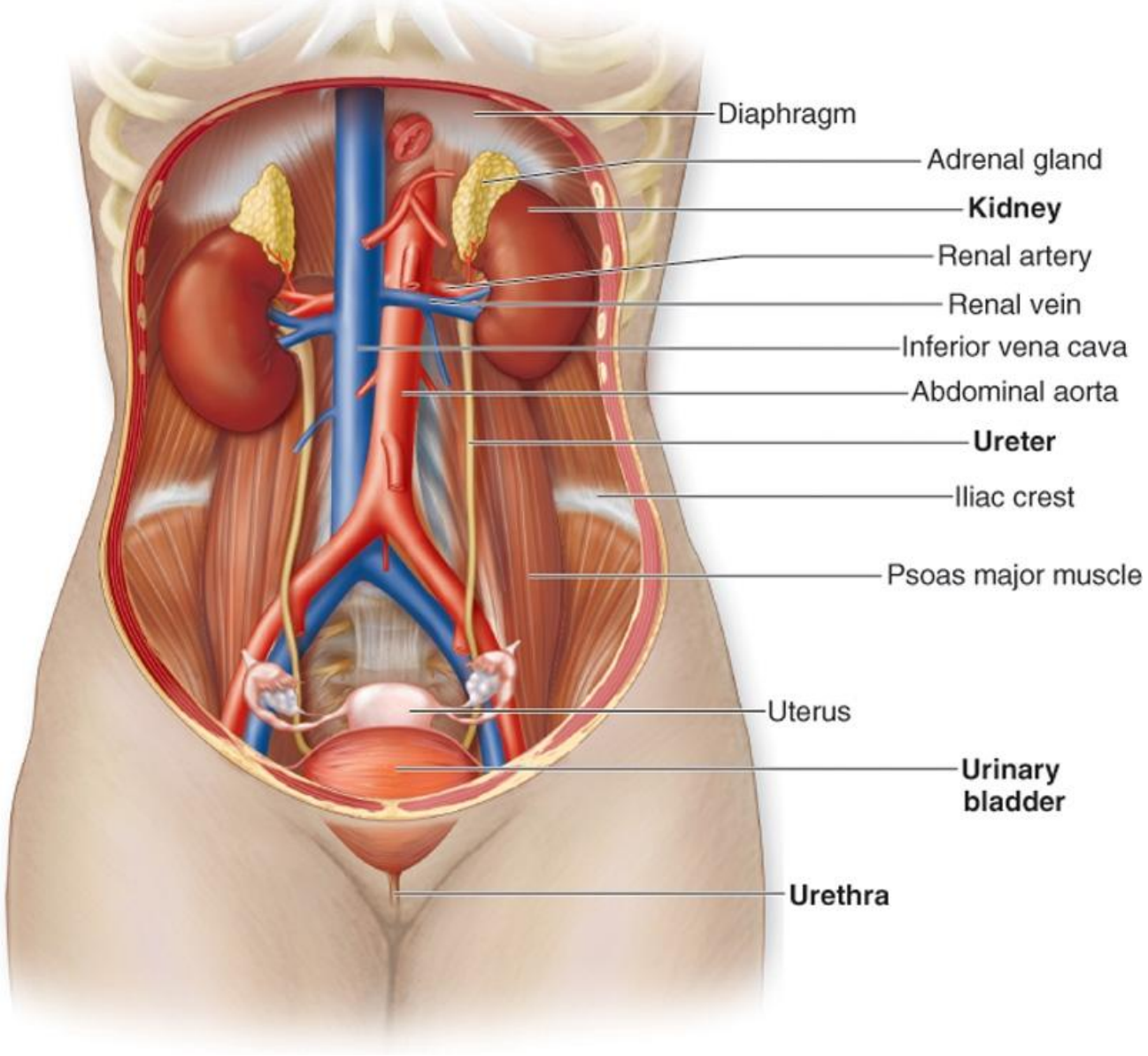
Ureters functions as tubes that actively convey urine from the kidneys to the bladder.



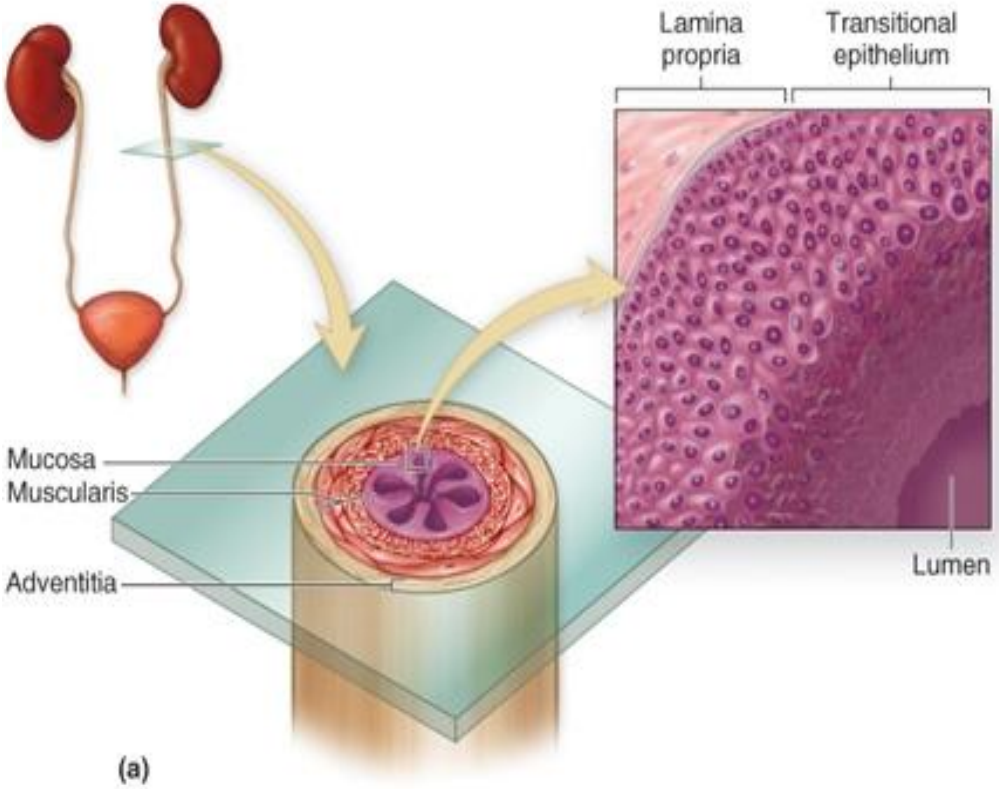
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URETERS

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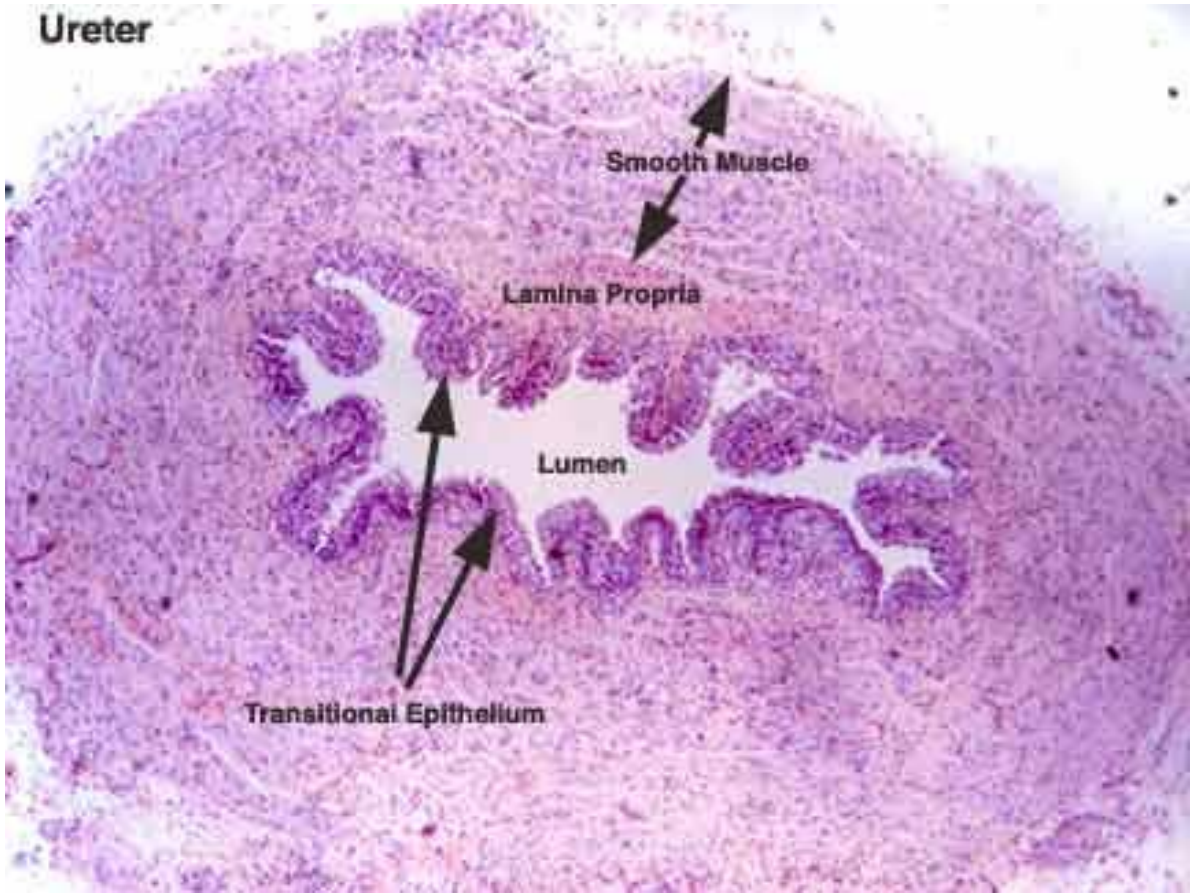
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(a)



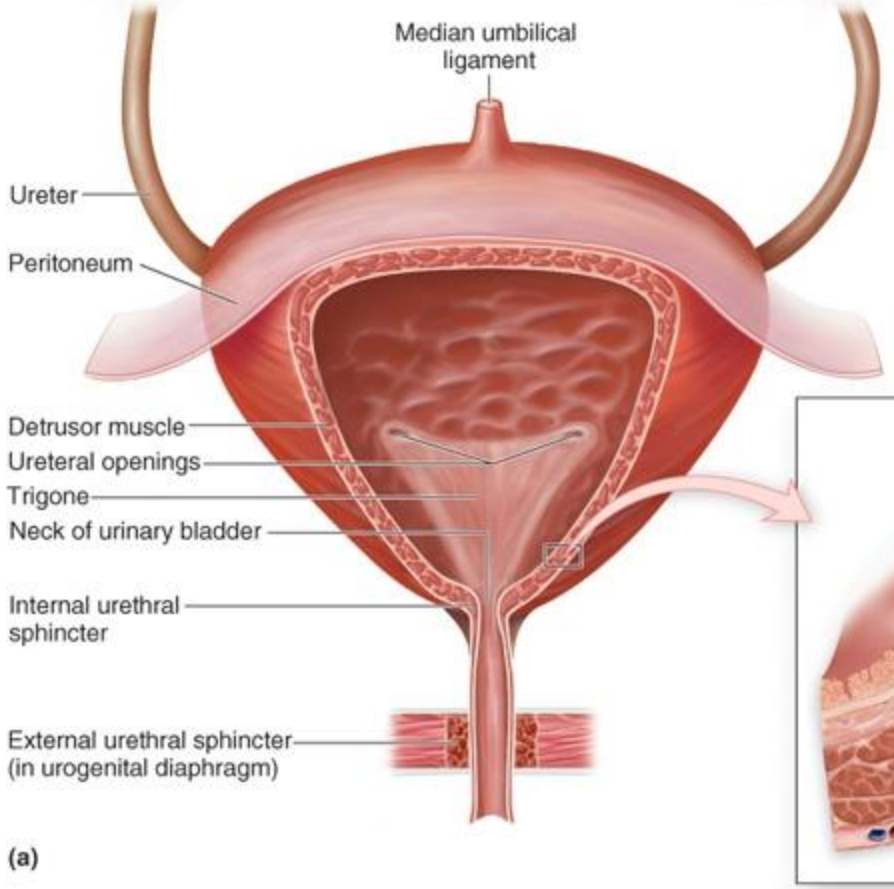
(b)

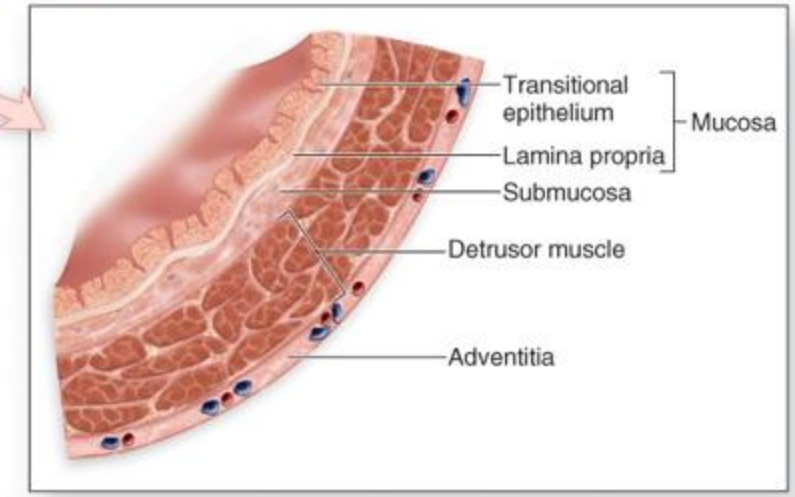

Ureter
Smooth Muscle
Lamina Propria
Lumen
Transitional Epithelium

URINARY BLADDER

Ex. ปวดท้องด้านซ้าย
= สิ่วด้านซ้ายมั้ย?

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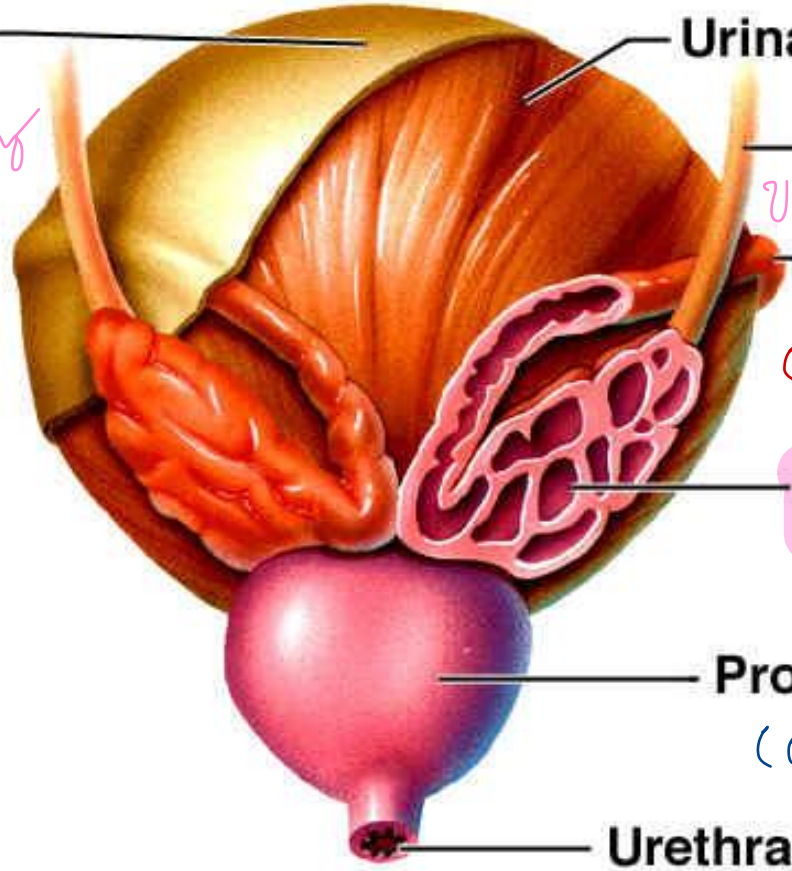


กระเพาะปัสสาวะ บรรจุน้ำได้ 200-250 ml
→ tran มากกว่านั้นได้

อยู่ช่องเชิงกราน

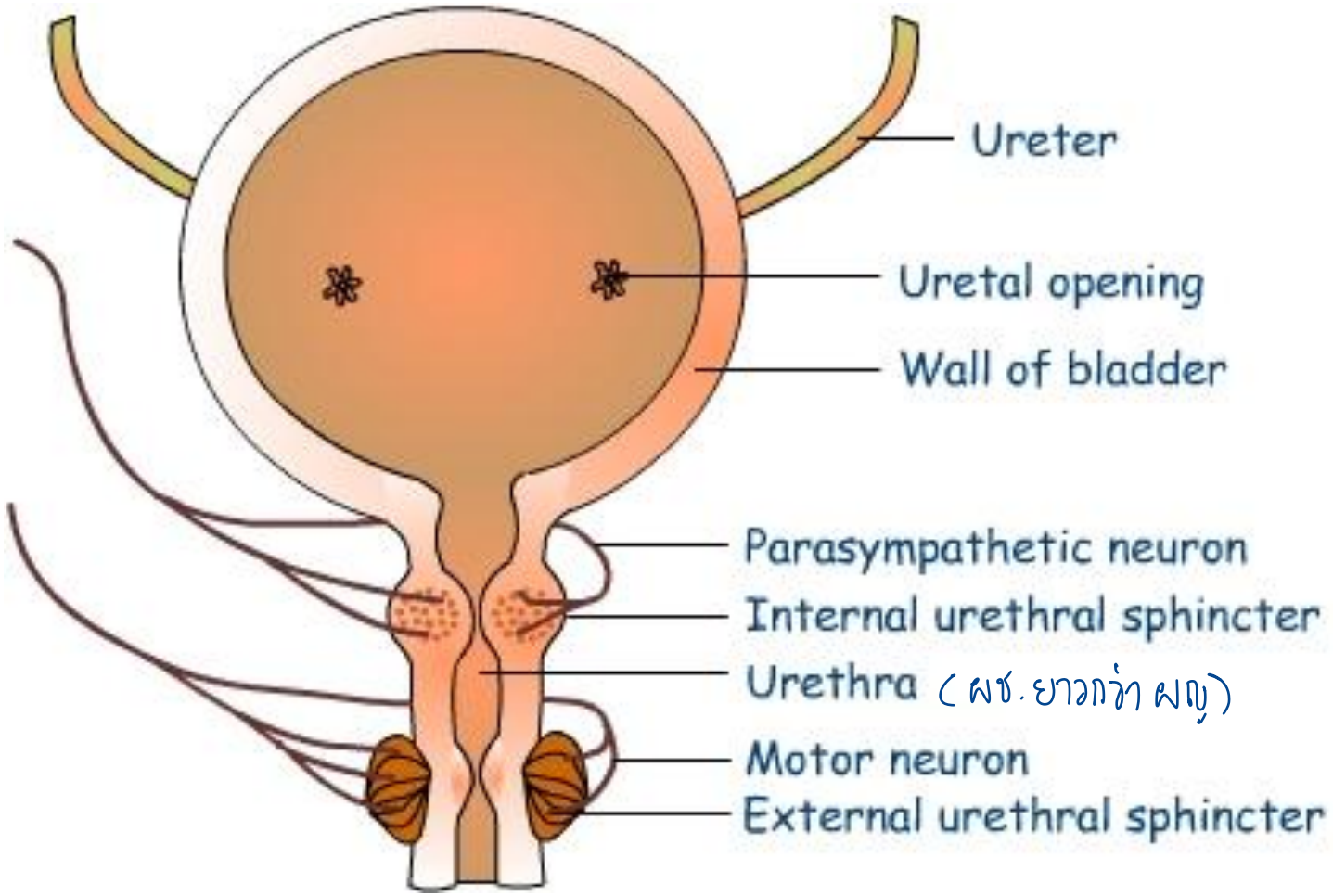
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Male Urinary Bladder—Posterior View (ด้านหลัง)



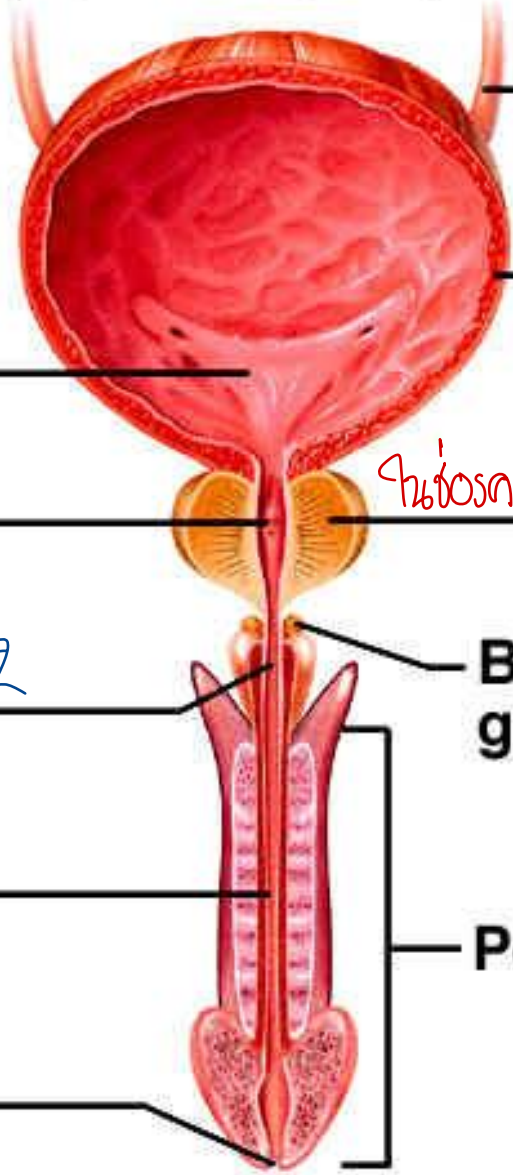
ก
ข
1
2 ท่อนำอสุจิ
(อัณฑะสร้าง)
3 เติมฟรุกโตส
4 เติมสปก.
(ต่อมลูกหมาก)

ผช. มี 2 รู
ป


Ureter
Uretal opening
Wall of bladder
Parasympathetic neuron
Internal urethral sphincter
Urethra (ผช. ยาวกว่า ผญ)
Motor neuron
External urethral sphincter

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Male Urinary Bladder and Urethra



Ureter

Urinary bladder

Trigone

1

Prostatic urethra

ในช่องคลอด

ปรับค. เป็นกรอก

Prostate gland (ใส่สปก.)

Urethra มี 3 ส่วน

2

Membranous urethra

Bulbourethral gland (น้ำหล่อลื่น)

3

Penile urethra

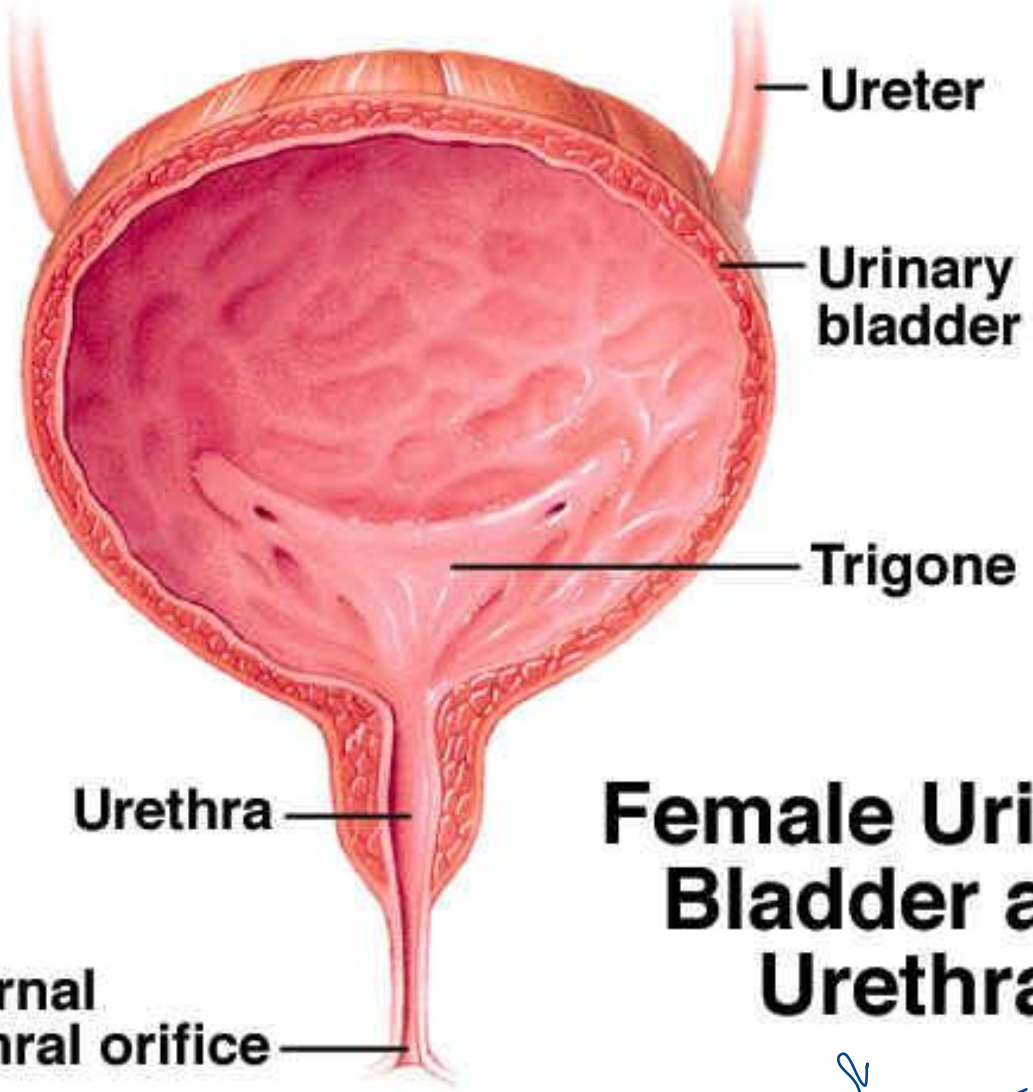
Penis

4

External urethral orifice

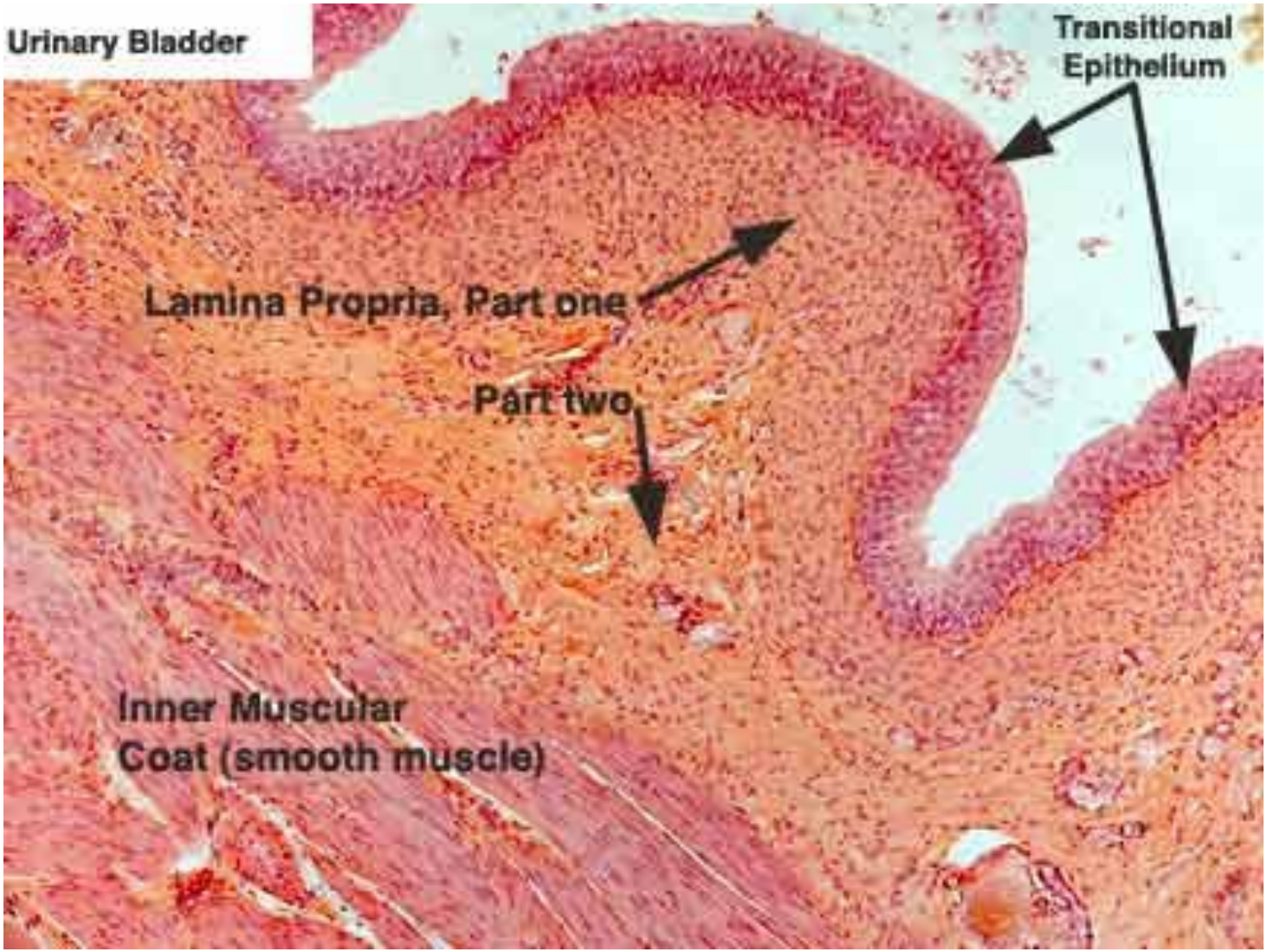
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ช่องทวาร

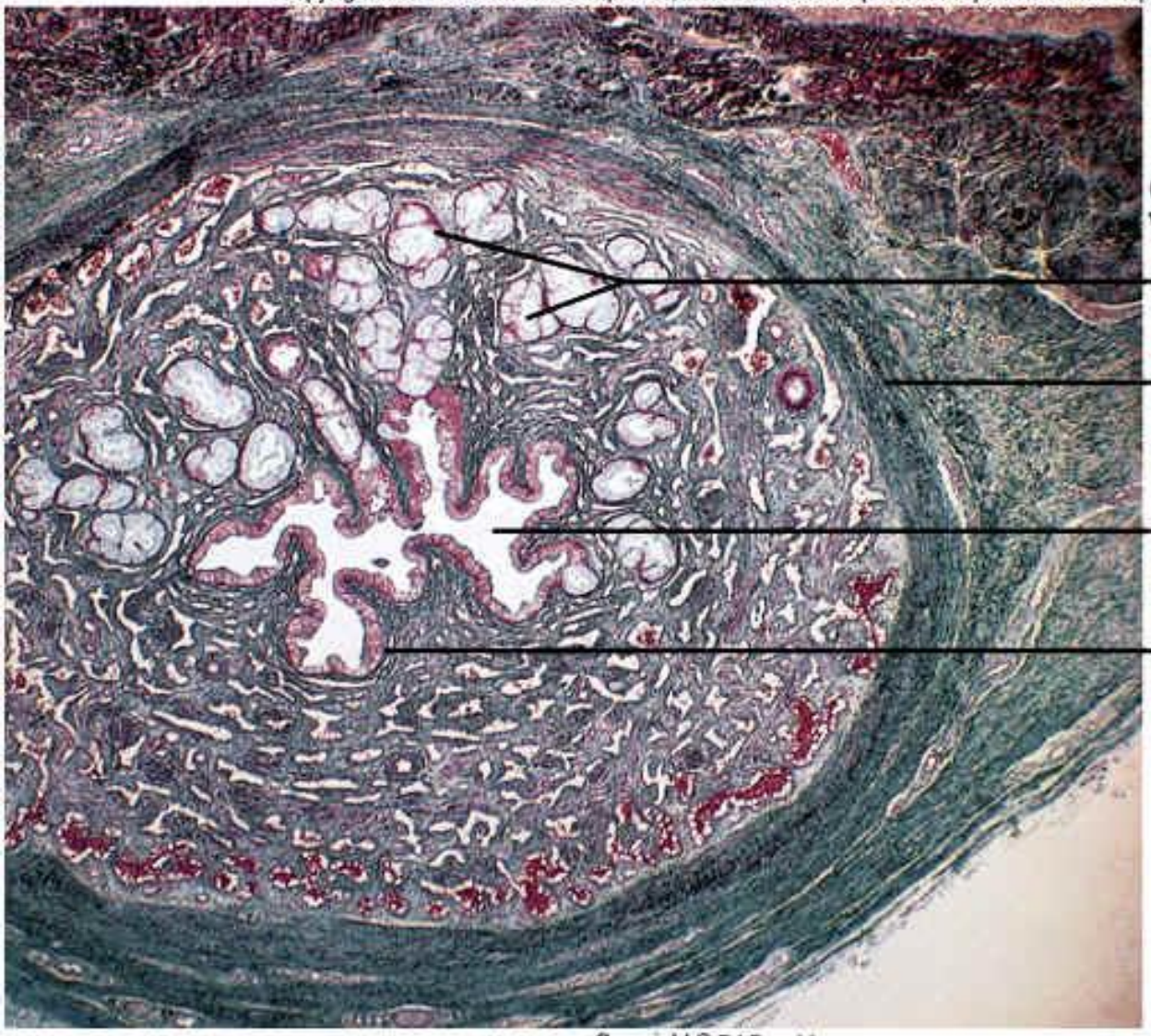


Female Urinary Bladder and Urethra

ผญ. มี 3 รู (ฉี่, ช่องคลอด ทวาร)


Urinary Bladder
Transitional Epithelium
Lamina Propria, Part one
Part two
Inner Muscular Coat (smooth muscle)

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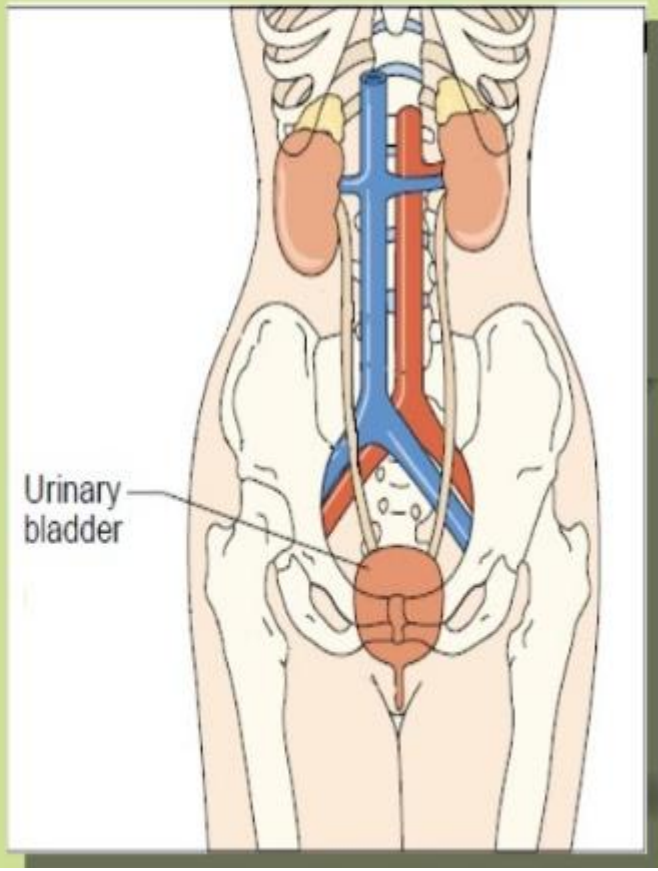


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Bladder

The urinary bladder is a muscular, hollow sac located just behind the pubic bone.

The bladder is characterized by its central, hollow area called the vesicle, which has two inlets (the ureters) and one outlet (the urethrovesical junction), which is surrounded by the bladder neck.



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Bladder

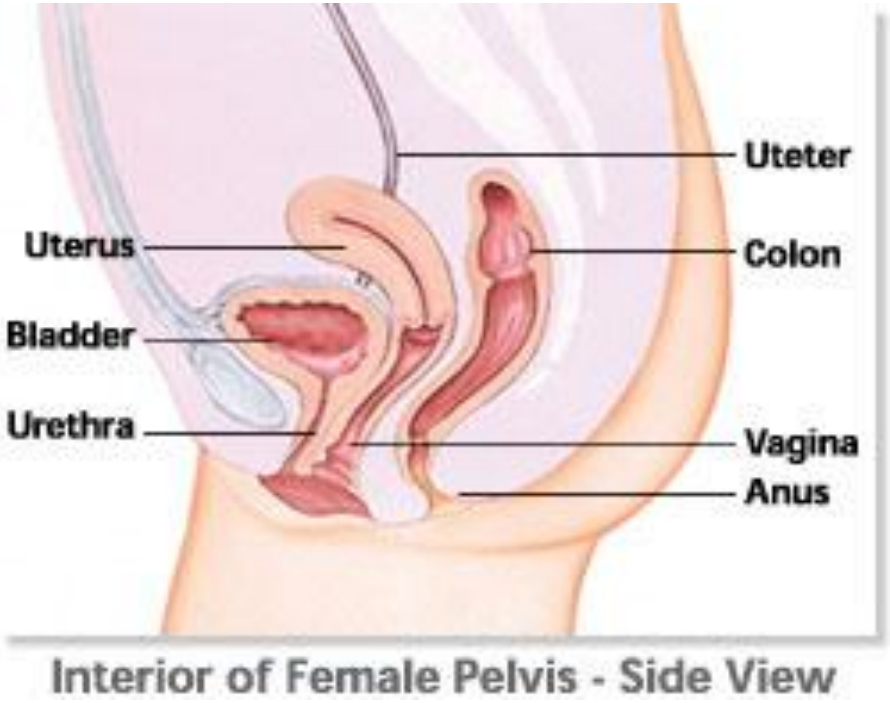
The wall of the bladder comprises four layers:

- adventitia
- detrusor
- lamina propria
- urothelium

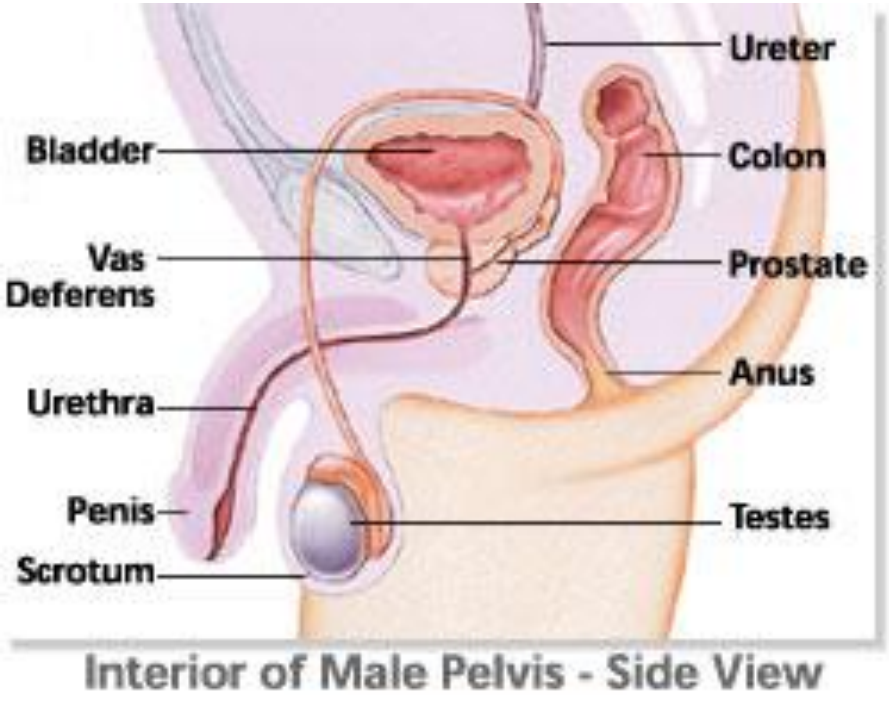


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Bladder

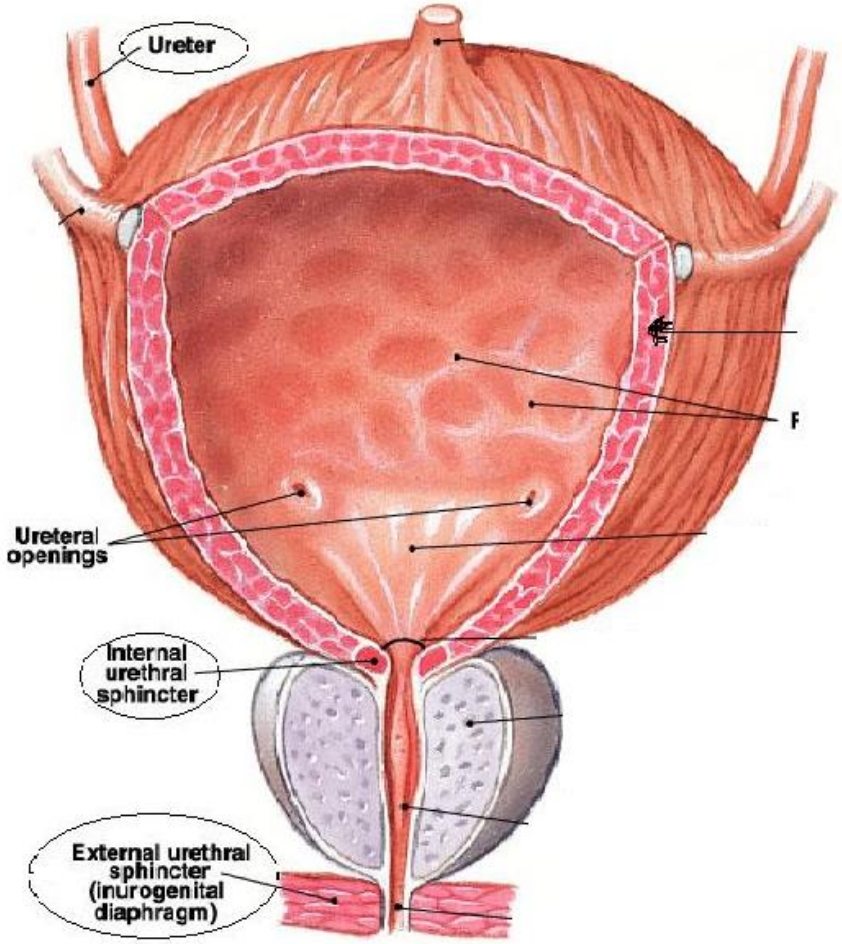


Interior of Female Pelvis - Side View



Interior of Male Pelvis - Side View

Bladder- structure of

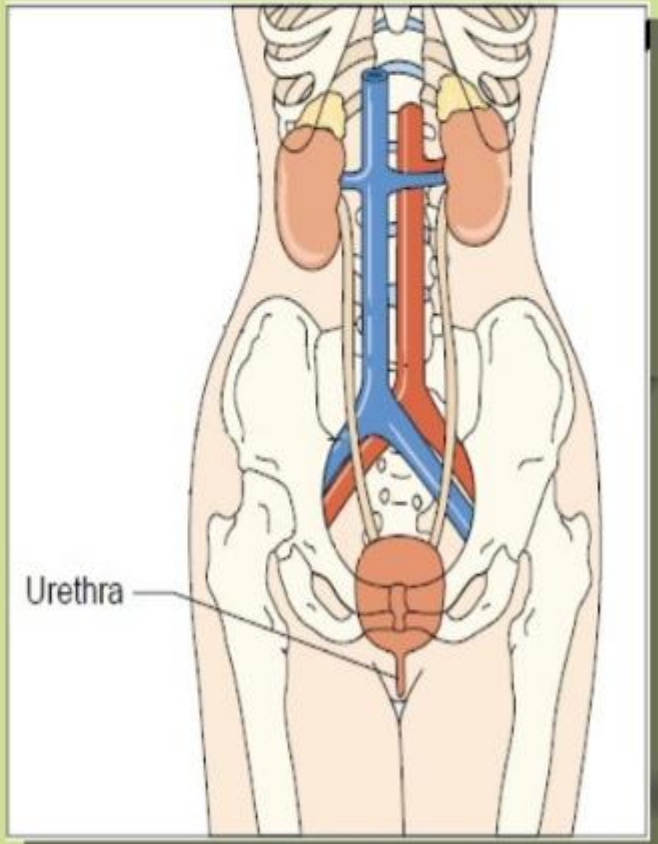


- 3 layers
  - Outer layer
    - Loose connective tissue
  - Middle layer
    - Smooth muscle and elastic fibres
  - Inner layer
    - Lined with transitional epithelium

Urethra

The urethra arises from the base of the bladder: In the male, it passes through the penis; in the female, it opens just anterior to the vagina.

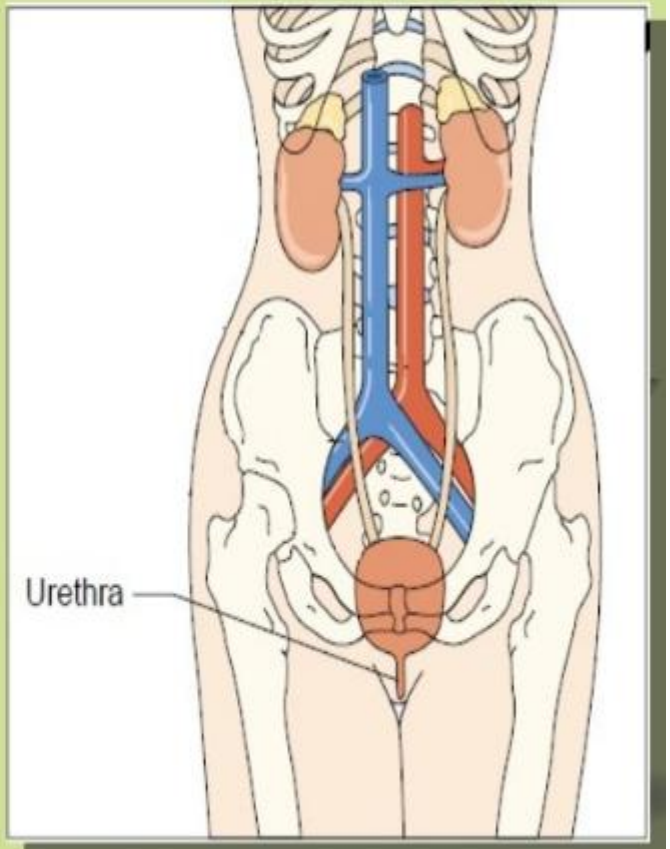
In the male, the prostate gland, which lies just below the bladder neck, surrounds the urethra posteriorly and laterally.



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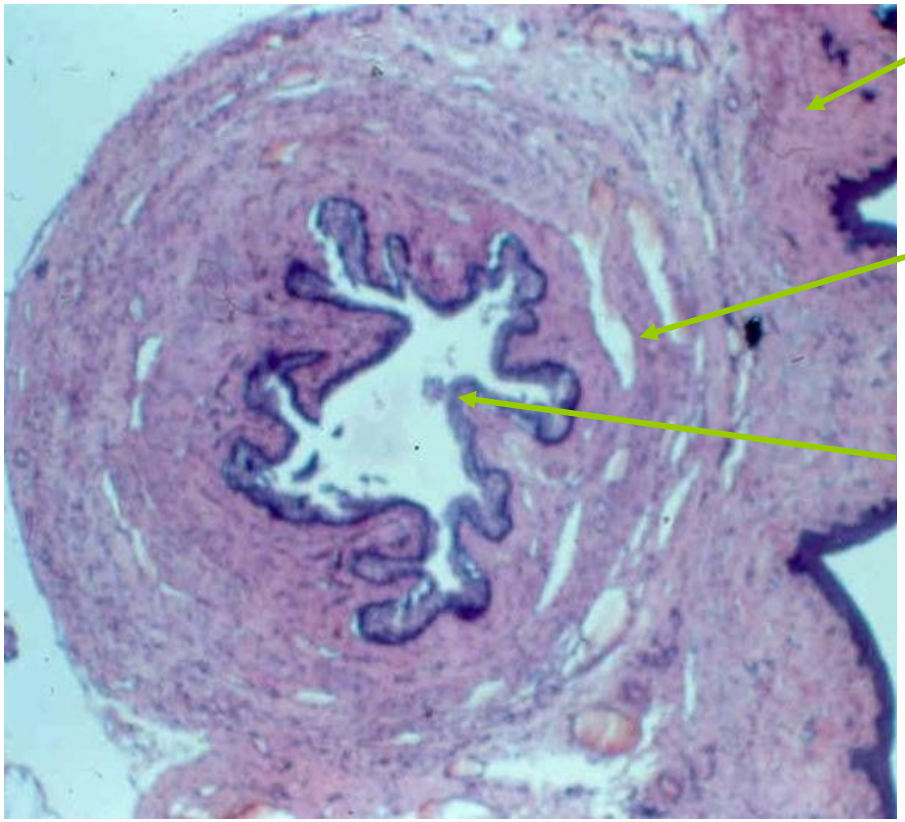
Urethra

The urethra is a muscular tube that drains urine from the body; it is 3–4 cm long in females, but closer to 20 cm in males.



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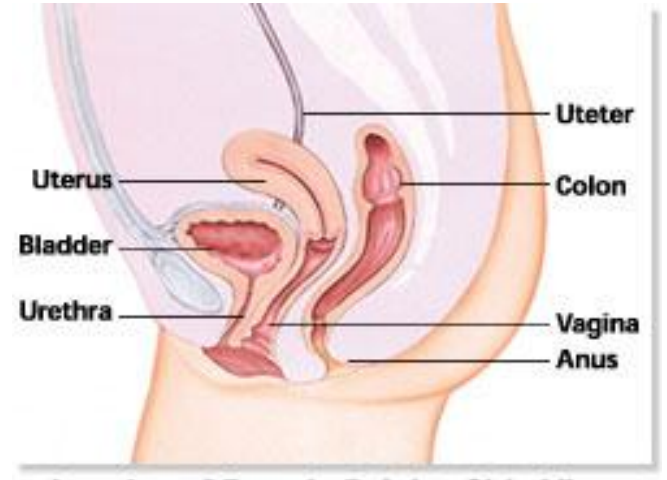
Urethra- structure of



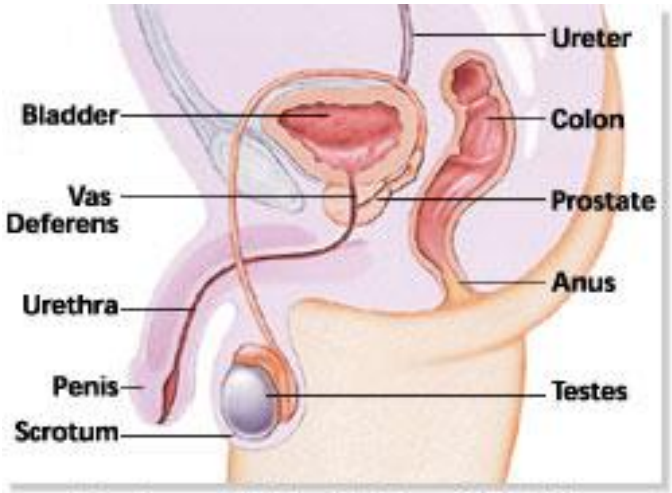
- Muscle layer
- Submucosa layer
- Mucosa

Urethra

- Extends from the base of the bladder to the outside world.
- Anatomical differences mean that male and female urethras are different.
  - Female: 4 cm long
  - Male: 14 cm long



Interior of Female Pelvis - Side View



Interior of Male Pelvis - Side View

Urethra

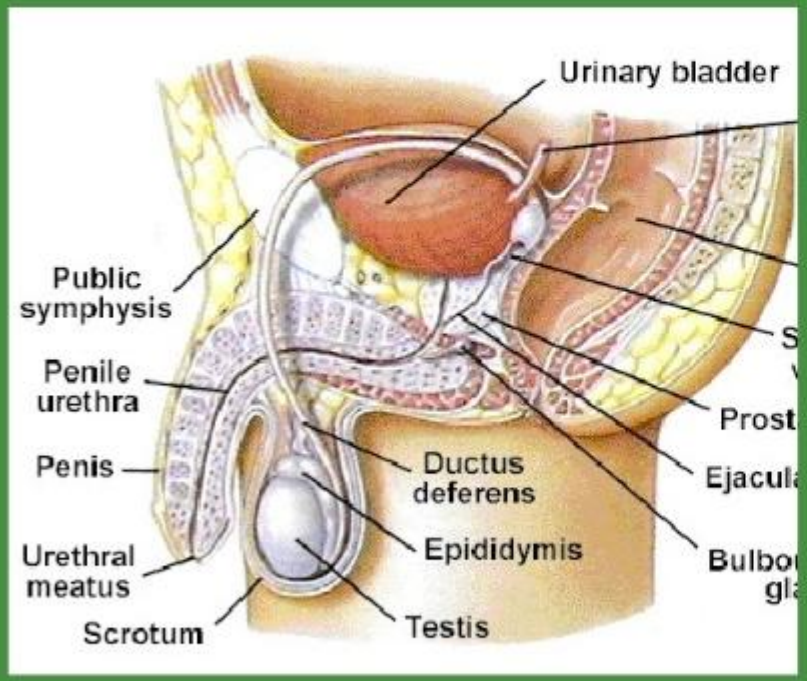
External urethral sphincters – voluntary at pelvic floor

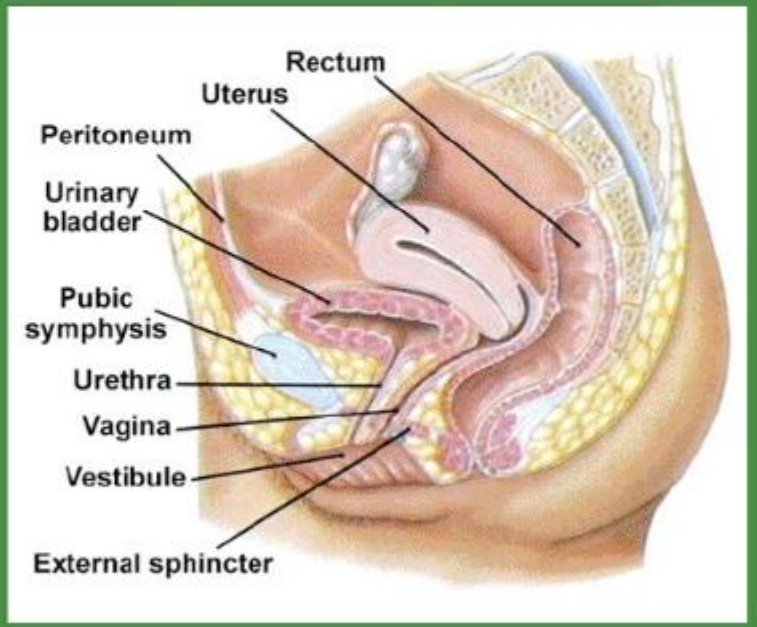
Female - short – from base of bladder to vestibule

Male

1. prostatic urethra – from base of bladder through prostate gland
2. membranous urethra – between prostate gland & base of penis
3. penile (spongy) urethra – traverses penis to orifice

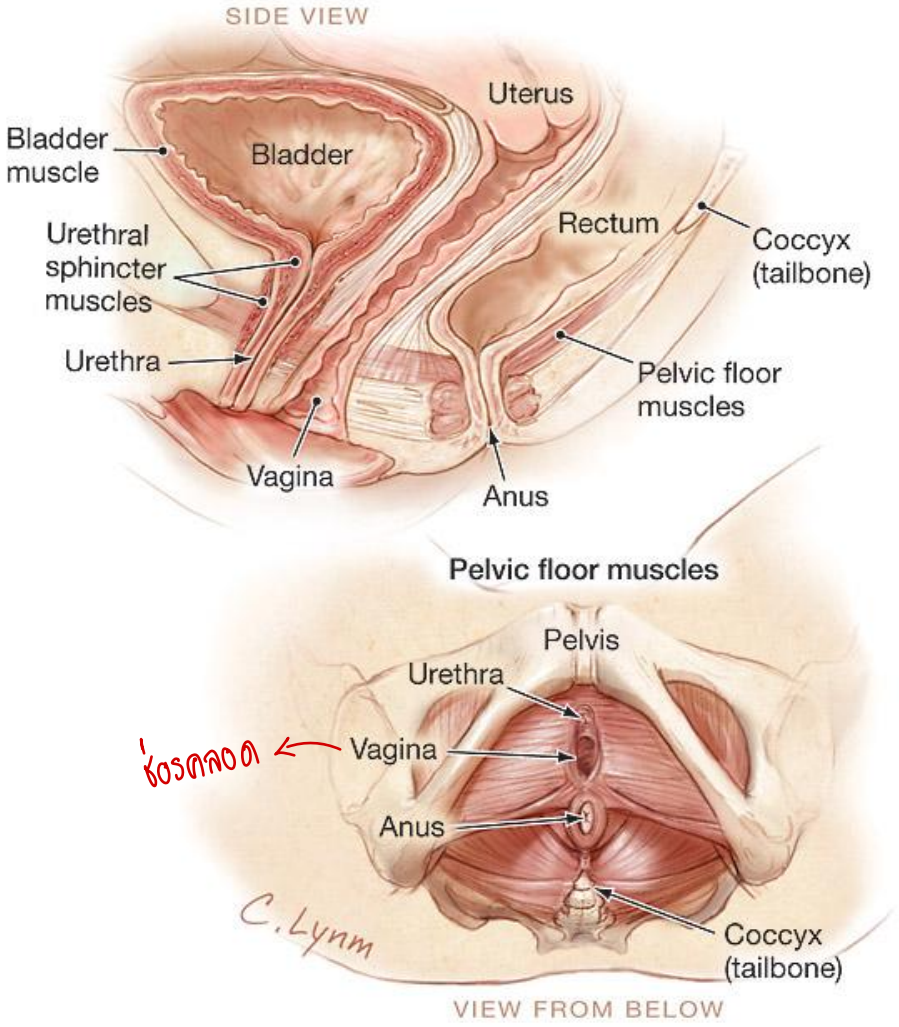
Male versus Female

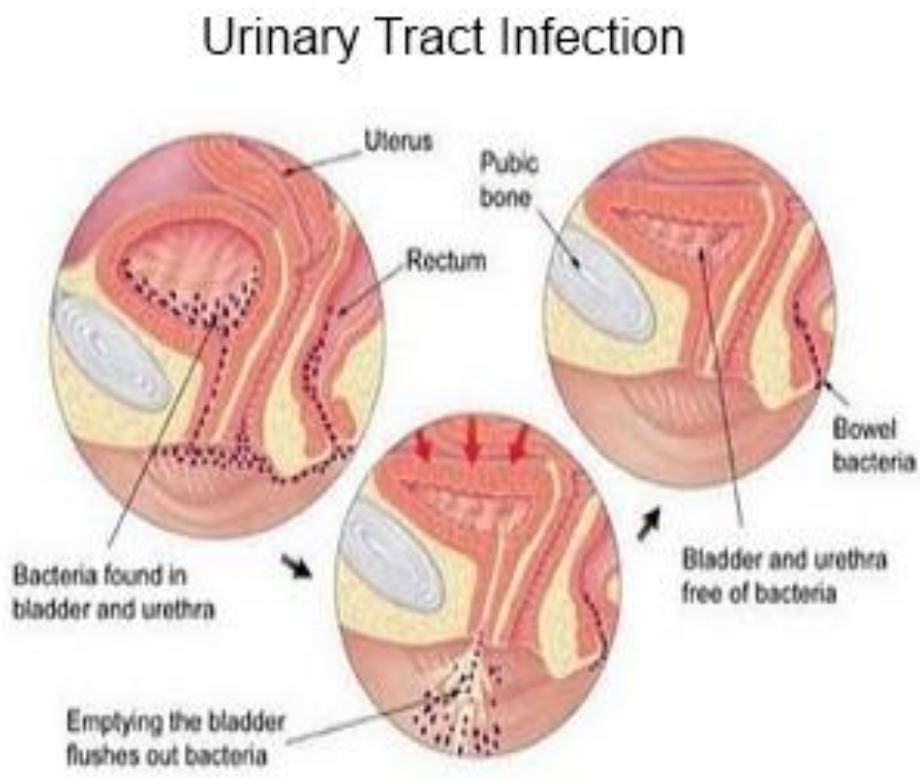


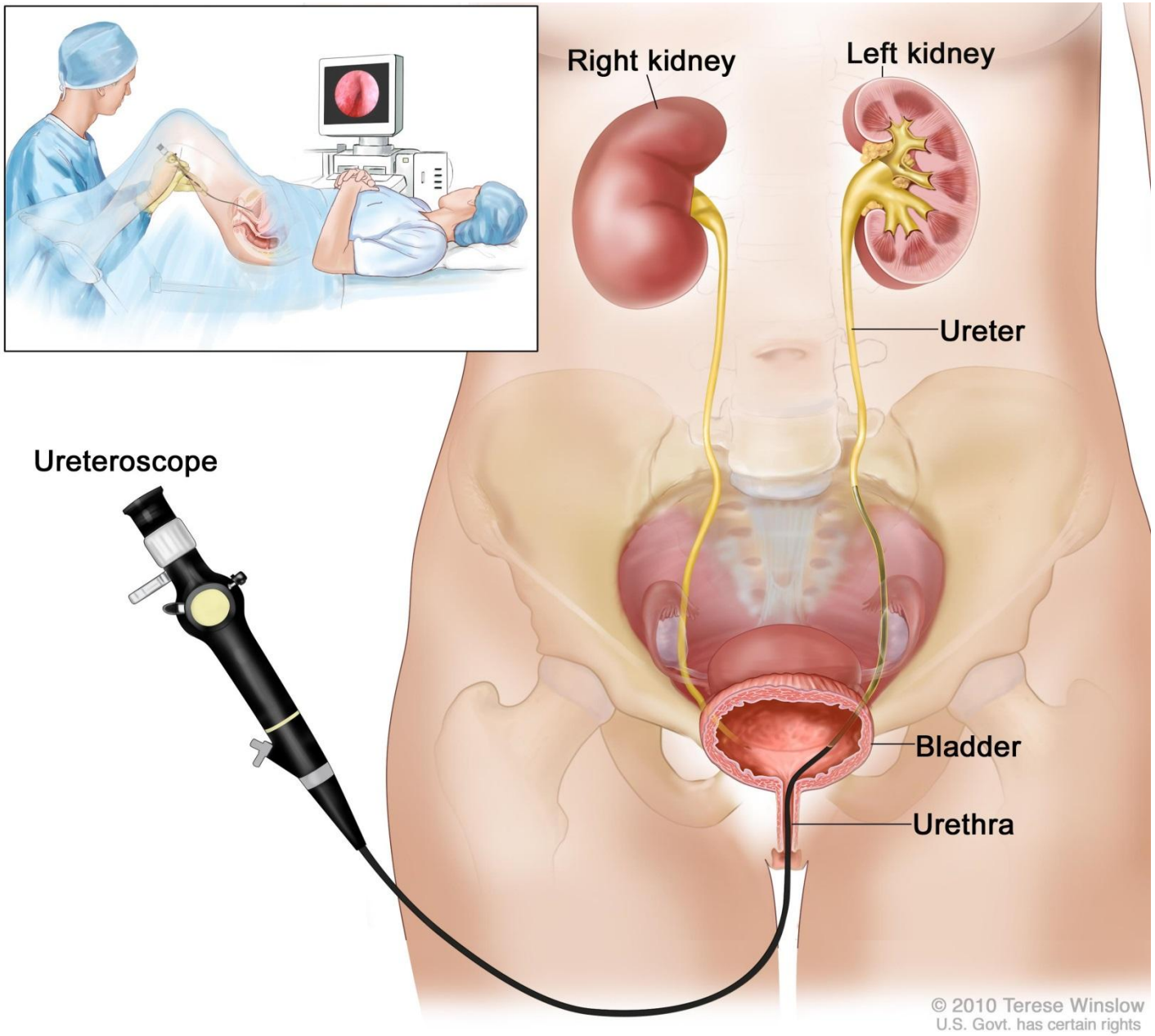


UTIs (esp. *E.coli*)

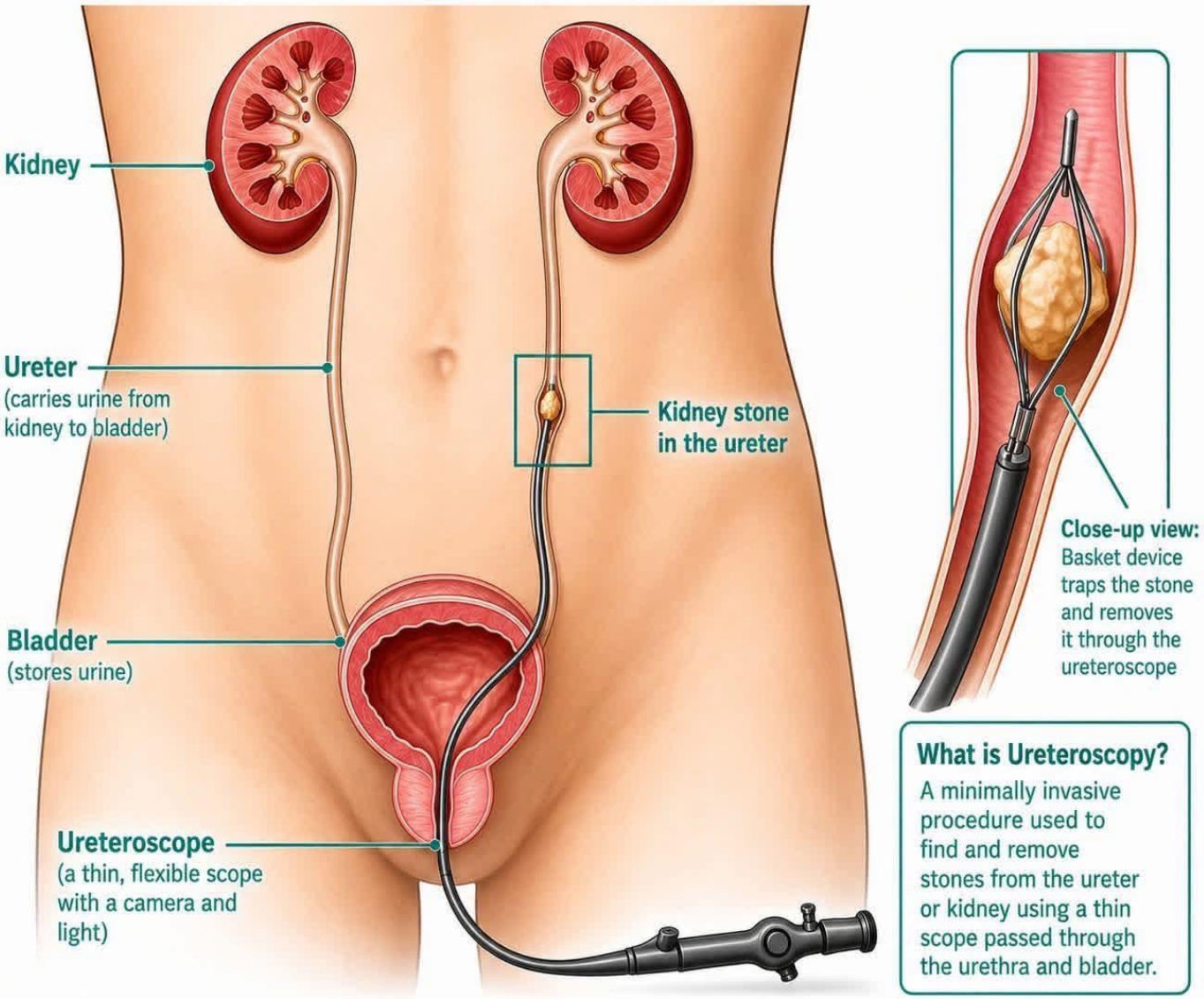
Female urethra






Right kidney
Left kidney
Ureter
Ureteroscope
Bladder
Urethra
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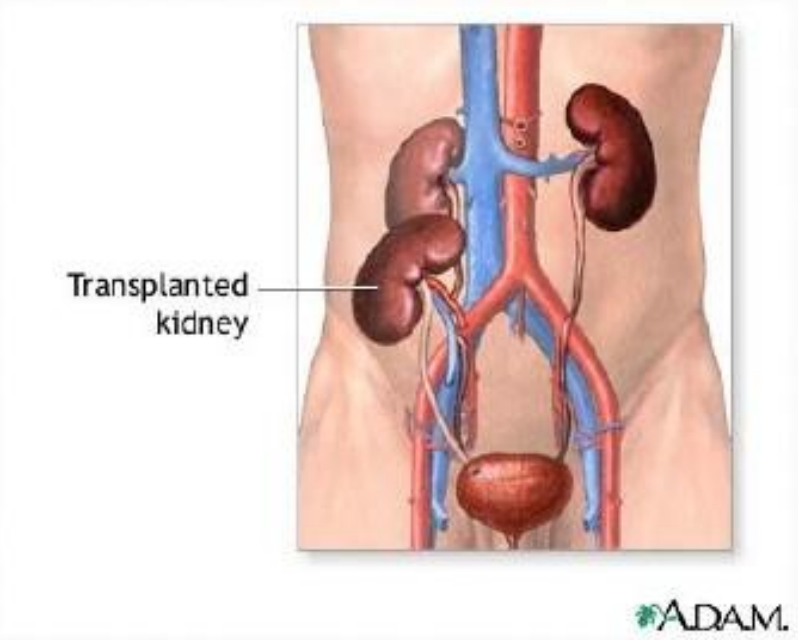
URETEROSCOPY FOR KIDNEY STONE REMOVAL

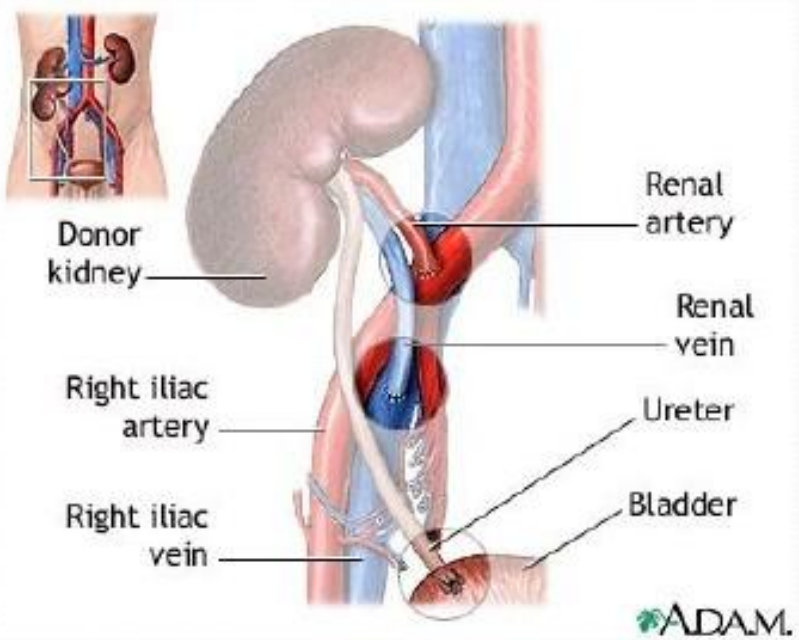


What is Ureteroscopy?

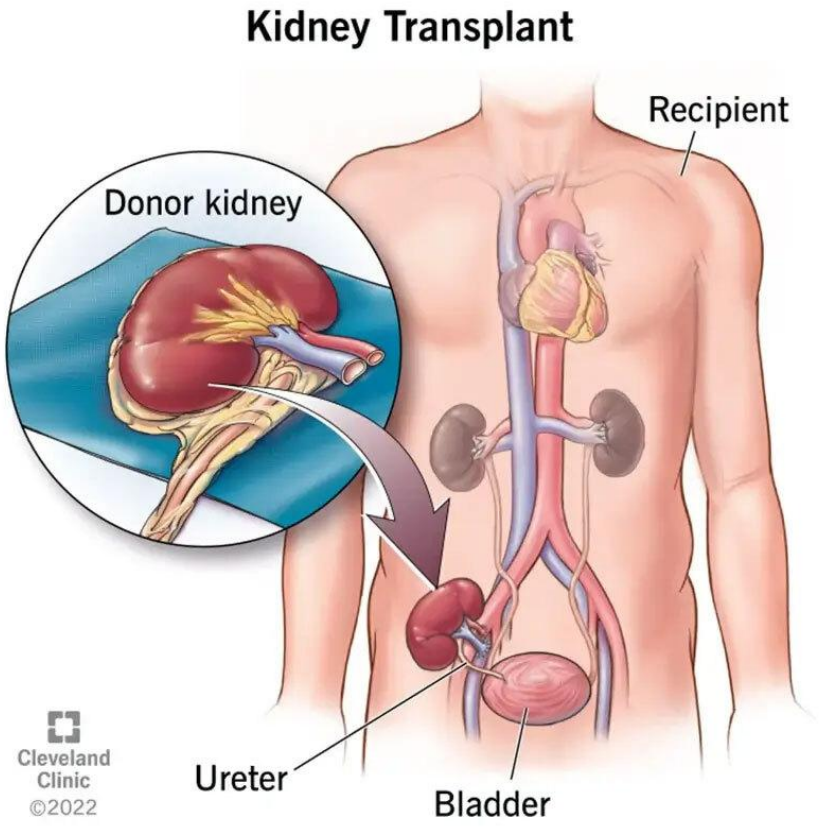
A minimally invasive procedure used to find and remove stones from the ureter or kidney using a thin scope passed through the urethra and bladder.

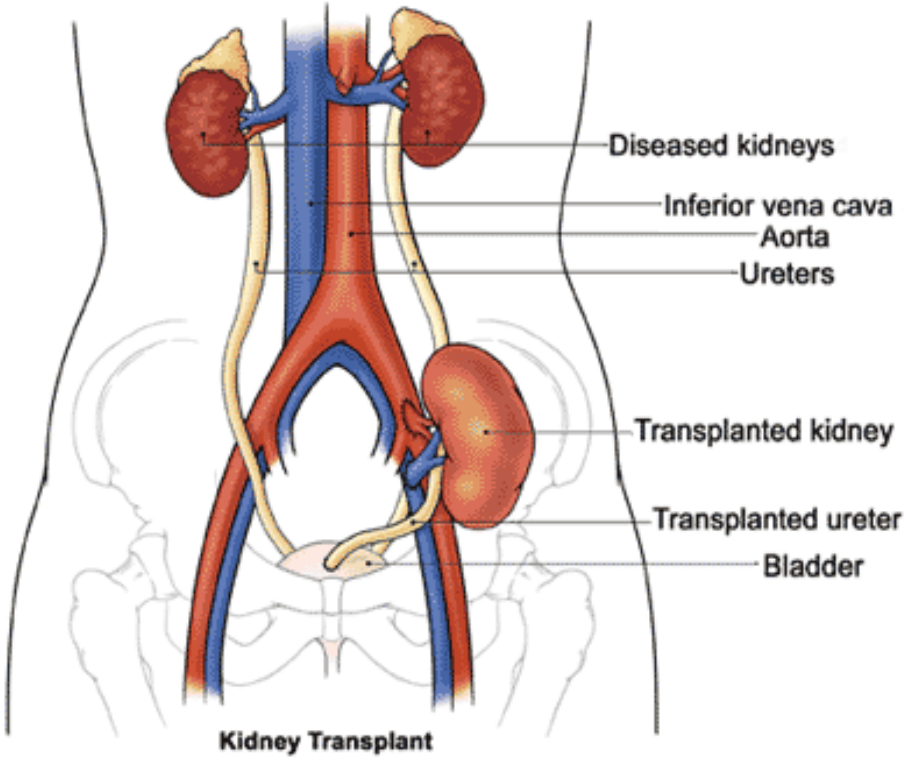
Anatomy of Renal Transplantation





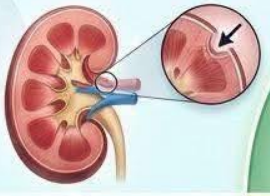
Kidney transplant





KIDNEY FAILURE STAGES

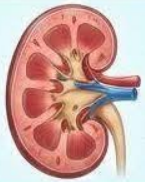
Chronic Kidney Disease (CKD) Progression Guide



Stage 1 CKD

eGFR: ≥ 90 ml/min/1.73m²

- Normal kidney function
- Possible protein in urine
- Usually no symptoms



Stage 2 CKD

eGFR: 60–89 ml/min

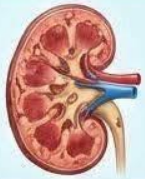
- Mild decline in kidney filtration
- Often asymptomatic
- Early lab abnormalities



Stage 3 CKD

eGFR: 30–59 ml/min

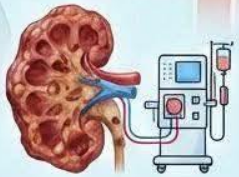
- Fatigue
- Swelling in legs
- Changes in urination
- Mild anemia



Stage 4 CKD

eGFR: 15–29 ml/min

- Severe fatigue
- Fluid retention
- Electrolyte imbalance
- High blood pressure



Stage 5 CKD (End Stage Renal Diseass)

eGFR: <15 ml/min

- Kidney unable to maintain body balance
- Dialysis or kidney transplant required

Common risk factors for CKD:



Diabetes mellitus



Hypertension



Chronic glomerulonephritis



Polycystic kidney disease



Long-term nephrotoxic drugs

Warning signs of advanced kidney disease:



Persistent swelling



Shortness of breath



Severe fatigue



Urine problem

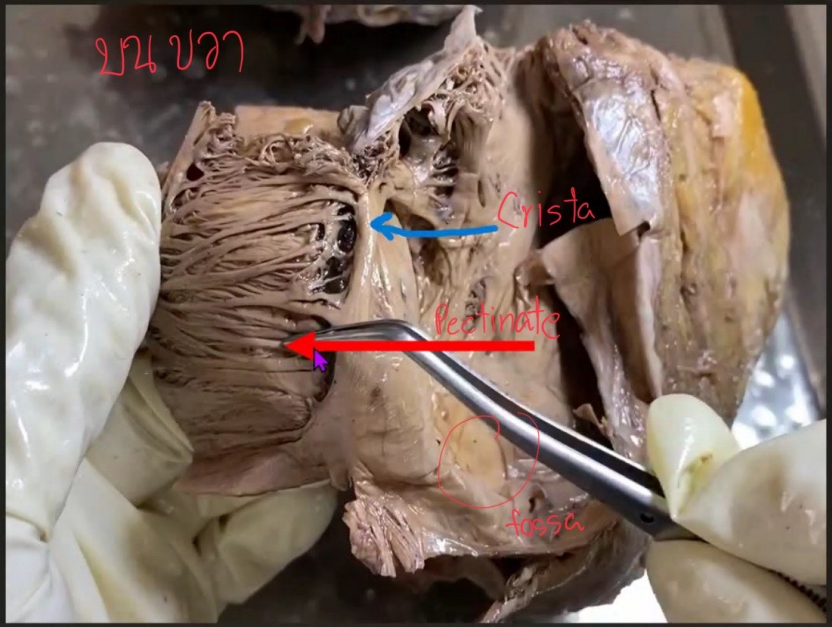


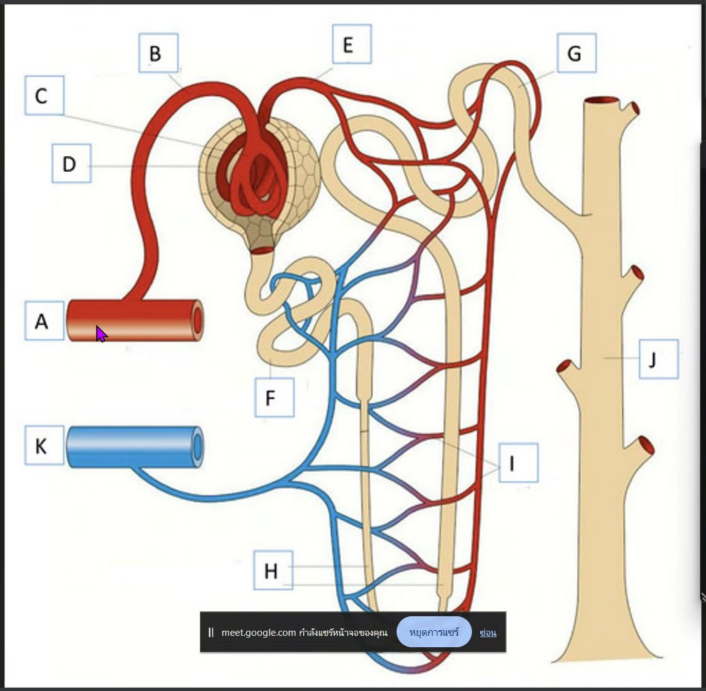
Reduced urine output

Educational nephrology infographic. Not intended for diagnosis or treatment decisions.

FEED ONLY

 @thereal.arjun18  @Arjun kharadi  @MedNurseAcademy97

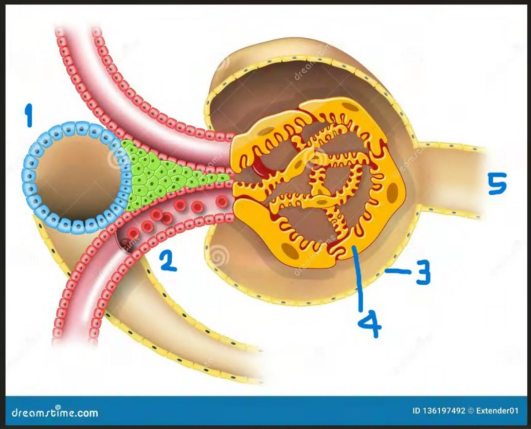

บน ขวา
Crista
Pectinate
fossa



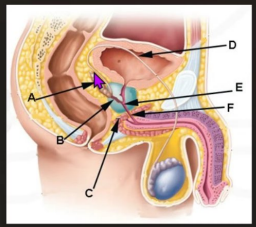
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